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Semiconductor module

Abstract

A semiconductor module includes a supporting substrate, a conductive substrate supported by the supporting substrate, a conductive bonding member provided between the supporting substrate and the conductive substrate, and a semiconductor element electrically bonded to an obverse surface of the conductive substrate and having a switching function. The conductive bonding member includes a metal base layer, a first layer, and a second layer. The first layer is provided between the base layer and the conductive substrate, and is in direct contact with the conductive substrate. The second layer is provided between the base layer and the supporting substrate, and is in direct contact with the supporting substrate.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a semiconductor module.

BACKGROUND ART

(2) Conventionally, semiconductor modules including power switching elements, such as metal oxide semiconductor field effect transistors (MOSFETs) or insulated gate bipolar transistors (IGBTs), have been known. The semiconductor modules are used in various electronic devices, such as industrial devices, home appliances, information terminals, and automobile devices. Patent Document 1 discloses a conventional semiconductor module (power module). The semiconductor module described in Patent Document 1 includes a semiconductor element and a supporting substrate (ceramic substrate). The semiconductor element is an IGBT made of silicon (Si), for example. The supporting substrate supports the semiconductor element. The supporting substrate includes an insulating base member and conductor layers stacked on the respective surfaces of the base member. The base member is made of ceramic, for example. Each of the conductor layers is made of copper (Cu), for example, and the semiconductor element is bonded to one of the conductor layers.

PRIOR ART DOCUMENT

Patent Document

(3) Patent Document 1: JP-A-2015-220382

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(4) In recent years, there has been a demand for energy-saving, sophisticated, and smaller electronic devices. In order to meet the demand, it is necessary to improve the performance and reduce size of semiconductor modules mounted on the electronic devices.

(5) The present disclosure is conceived in view of the above circumstances, and aims to provide a semiconductor module having a module configuration preferable for improving the heat dissipation property and suppressing the peeling between the conductive substrate and the supporting

substrate.

(6) Means to Solve the Problem

(7) A semiconductor module according to the present disclosure includes a supporting substrate, a conductive substrate, a first conductive bonding member, and a semiconductor element. The supporting substrate has a supporting surface that faces in one sense of a thickness direction. The conductive substrate has an obverse surface and a reverse surface that face away from each other in the thickness direction, and is supported by the supporting substrate such that the reverse surface faces the supporting surface. The first conductive bonding member is provided between the supporting substrate and the conductive substrate, and electrically bonds the supporting substrate and the conductive substrate. The semiconductor element is electrically bonded to the obverse surface, and has a switching function. The first conductive bonding member has a first base layer made of metal, a first layer, and a second layer. The first layer is provided between the first base layer and the conductive substrate and bonded in direct contact with the conductive substrate at a bonding interface with the conductive substrate. The second layer is provided between the first base layer and the supporting substrate and bonded in direct contact with the supporting substrate at a bonding interface with the supporting substrate.

Advantages of the Invention

(8) The configuration as described above can provide a module configuration preferable for improving the heat dissipation property and suppressing the peeling between the conductive substrate and the supporting substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a semiconductor module according to a first embodiment.

(2) FIG. 2 is a perspective view corresponding to FIG. 1 but omitting a sealing resin, a resin member, and a resin-filling portion.

(3) FIG. 3 is a perspective view corresponding to FIG. 2 but omitting a conducting member.

(4) FIG. 4 is a plan view illustrating the semiconductor module according to the first embodiment.

(5) FIG. 5 is a plan view corresponding to FIG. 4 but showing the sealing resin, the resin member, and the resin-filling portion with imaginary lines.

(6) FIG. 6 is a partially enlarged view showing a part of FIG. 5 but omitting the imaginary lines of the sealing resin, the resin member, and the resin-filling portion.

(7) FIG. 7 is a partially enlarged view showing a part of FIG. 6.

(8) FIG. 8 is a plan view corresponding to FIG. 5 but showing a part of the conducting member with an imaginary line.

(9) FIG. 9 is a front view illustrating the semiconductor module according to the first embodiment.

(10) FIG. 10 is a bottom view illustrating the semiconductor module according to the first embodiment.

(11) FIG. 11 is a left side view illustrating the semiconductor module according to the first embodiment.

(12) FIG. 12 is a right side view illustrating the semiconductor module according to the first embodiment.

(13) FIG. 13 is a cross-sectional view taken along line XIII-XIII of FIG. 5.

(14) FIG. 14 is a cross-sectional view taken along line XIV-XIV of FIG. 5.

(15) FIG. 15 is a partially enlarged view showing a part of FIG. 14.

(16) FIG. 16 is a cross-sectional view taken along line XVI-XVI of FIG. 5.

(17) FIG. 17 is a cross-sectional view taken along line XVII-XVII of FIG. 5.

- (18) FIG. 18 is a cross-sectional view taken along line XVIII-XVIII of FIG. 5.
- (19) FIG. 19 is a cross-sectional view taken along line XIX-XIX of FIG. 5.
- (20) FIG. 20 shows an example of the circuit configuration of the semiconductor module according to the first embodiment.
- (21) FIG. 21 is a plan view illustrating a step of a method for manufacturing the semiconductor module according to the first embodiment.
- (22) FIG. 22 is a schematic cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment.
- (23) FIG. 23 is a plan view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment.
- (24) FIG. 24 is a cut end view illustrating a step of the manufacturing method according to the first embodiment, and corresponds to the cross-sectional view of FIG. 13.
- (25) FIG. 25 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment, and corresponds to an enlarged view of a part of the cross-section in FIG. 13.
- (26) FIG. 26 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment, and corresponds to an enlarged view of a part of the cross-section in FIG. 14.
- (27) FIG. 27 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment, and corresponds to an enlarged view of a part of the cross-section in FIG. 14.
- (28) FIG. 28 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment, and corresponds to an enlarged view of a part of the cross-section in FIG. 13.
- (29) FIG. 29 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module according to the first embodiment, and corresponds to an enlarged view of a part of the cross-section in FIG. 14.
- (30) FIG. 30 is a plan view similar to FIG. 5, illustrating a semiconductor module according to a second embodiment.
- (31) FIG. 31 is a partially enlarged view showing a part of FIG. 30 but omitting the imaginary lines of the sealing resin, the resin member, and the resin-filling portion.
- (32) FIG. 32 is a partially enlarged view showing a part of FIG. 31.
- (33) FIG. 33 is a plan view similar to FIG. 5, illustrating a semiconductor module according to a third embodiment.

(34) FIG. 34 is a cross-sectional view taken along line XXXIV-XXXIV of FIG. 33.

MODE FOR CARRYING OUT THE INVENTION

- (35) The following describes preferred embodiments of a semiconductor module according to the present disclosure with reference to the drawings. In the following, the same or similar elements are provided with the same reference signs, and redundant descriptions are omitted.
- (36) FIGS. 1 to 20 illustrate a semiconductor module A1 according to a first embodiment. The semiconductor module A1 includes a plurality of semiconductor elements 10, a conductive substrate 2, a supporting substrate 3, a plurality of input terminals 41 to 43, a plurality of output terminals 44, a plurality of control terminals 45, a control terminal support 5, a conducting member 6, a first conductive bonding member 71, a second conductive bonding member 72, a plurality of wires 731 to 735, a sealing resin 8, resin members 87, and resin-filling portions 88.
- (37) FIG. 1 is a perspective view illustrating the semiconductor module A1. FIG. 2 is a perspective view corresponding to FIG. 1 but omitting the sealing resin 8, the resin members 87, and the resin-filling portions 88. FIG. 3 is a perspective view corresponding to FIG. 2 but omitting the conducting member 6. FIG. 4 is a plan view illustrating the semiconductor module A1. FIG. 5 is a plan view corresponding to FIG. 4 but showing the sealing resin 8, the resin members 87, and the

resin-filling portions **88** with imaginary lines. FIG. **6** is a partially enlarged view showing a part of FIG. **5**. In FIG. **6**, the imaginary lines of the sealing resin **8**, the resin members **87**, and the resin-filling portions **88** are omitted. FIG. **7** is a partially enlarged view showing a part of FIG. **6**. FIG. **8** is a plan view corresponding to FIG. **5** but showing a part of the conducting member **6** (a second conducting member **62** described below) with an imaginary line. FIG. **9** is a front view illustrating the semiconductor module **A1**. FIG. **10** is a bottom view illustrating the semiconductor module **A1**. FIG. **11** is a left side view illustrating the semiconductor module **A1**. FIG. **12** is a right side view illustrating the semiconductor module **A1**. FIG. **13** is a cross-sectional view taken along line XIII-XIII of FIG. **5**. FIG. **14** is a cross-sectional view taken along line XIV-XIV of FIG. **5**. FIG. **15** is a partially enlarged view showing a part of FIG. **14**. FIG. **16** is a cross-sectional view taken along line XVI-XVI of FIG. **5**. FIG. **17** is a cross-sectional view taken along line XVII-XVII of FIG. **5**. FIG. **18** is a cross-sectional view taken along line XVIII-XVIII of FIG. **5**. FIG. **19** is a cross-sectional view taken along line XIX-XIX of FIG. **5**. In FIGS. **2**, **3**, **7**, **14**, and **18**, the wires **731** to **735** are omitted. FIG. **20** shows an example of the circuit configuration of the semiconductor module **A1**. In the circuit diagram of FIG. **20**, only one of a plurality of first semiconductor elements **10A** (described below) and only one of a plurality of second semiconductor elements **10B** (described below) are illustrated, and the rest of the first semiconductor elements **10A** and the second semiconductor elements **10B** are omitted.

(38) For convenience, reference is made to three mutually perpendicular directions, i.e., x direction, y direction, and z direction. The z direction corresponds to the thickness direction of the semiconductor module **A1**. The x direction corresponds to the horizontal direction in the plan view (see FIG. **4**) of the semiconductor module **A1**. The y direction corresponds to the vertical direction in the plan view (see FIG. **4**) of the semiconductor module **A1**. One sense of the x direction is defined as x1 direction, and the other sense as x2 direction. Similarly, one sense of the y direction is defined as y1 direction, and the other sense as y2 direction. One sense of the z direction is defined as z1 direction, and the other sense as z2 direction. In the following description, a “plan view” is a view seen in the z direction. The z1 direction may be referred to as “downward”, and the z2 direction may be referred to as “upward”. The z direction is an example of the “thickness direction”, the x direction is an example of a “first direction”, and the y direction is an example of a “second direction”. In the following description, the mutually opposite senses in one direction are referred to as “one sense” and “the other sense”, respectively, but the present disclosure is not limited thereto. Specifically, the x2 direction is referred to as “one sense of the first direction”, and the x1 direction is referred to as “the other sense of the first direction”. Similarly, the y2 direction is referred to as “one sense of the second direction”, and the y1 direction is referred to as “the other sense of the second direction”. Furthermore, the z2 direction is referred to as “one sense of the thickness direction” and the z1 direction is referred to as “the other sense of the thickness direction”.

(39) The semiconductor elements **10** form the functional core of the semiconductor module **A1**. The semiconductor elements **10** are made of a semiconductor material that mainly contains silicon carbide (SiC), for example. The semiconductor material is not limited to SiC, and may be silicon (Si), gallium arsenide (GaAs) or gallium nitride (GaN). Each of the semiconductor elements **10** has a switching function unit **Q1** (see FIG. **20**) composed of a metal-oxide-semiconductor field-effect transistor (MOSFET), for example. The switching function unit **Q1** is not limited to a MOSFET, and may be another transistor, which is, for example, a field-effect transistor such as a metal-insulator-semiconductor FET or a bipolar transistor such as an IGBT. The semiconductor elements **10** are the same elements. The semiconductor elements **10** are n-channel MOSFETs, for example, but may be p-channel MOSFETs instead.

(40) As shown in FIG. **15**, each of the semiconductor elements **10** has an element obverse surface **101** and an element reverse surface **102**. The element obverse surface **101** and the element reverse surface **102** of each semiconductor element **10** are spaced apart from each other in the z direction.

The element obverse surface **101** faces in the z2 direction, and the element reverse surface **102** faces in the z1 direction.

(41) The semiconductor elements **10** include a plurality of first semiconductor elements **10A** and a plurality of second semiconductor elements **10B**. In the present embodiment, the semiconductor module **A1** includes three first semiconductor elements **10A** and three second semiconductor elements **10B**. However, the number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** are not limited thereto, and may be changed as appropriate according to the performance required for the semiconductor module **A1**. In the example shown in FIG. **8**, three first semiconductor elements **10A** and three second semiconductor elements **10B** are provided. The number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** may each be one, two, or no less than four. The number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** may be equal or different. The number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** are determined according to the current capacity handled by the semiconductor module **A1**.

(42) As shown in FIG. **20**, the semiconductor module **A1** may be configured as a half-bridge switching circuit. In this case, the first semiconductor elements **10A** constitute an upper arm circuit of the semiconductor module **A1**, and the second semiconductor elements **10B** constitute a lower arm circuit of the semiconductor module **A1**. In the upper arm circuit, the first semiconductor elements **10A** are connected in parallel to each other, and in the lower arm circuit, the second semiconductor elements **10B** are connected in parallel to each other. Each of the first semiconductor elements **10A** and each of the second semiconductor elements **10B** are connected in series to form a bridge layer.

(43) As shown in FIGS. **8** and **16**, for example, the first semiconductor elements **10A** are mounted on the conductive substrate **2**. In the example shown in FIG. **8**, the first semiconductor elements **10A** are aligned in the y direction and spaced apart from each other. The first semiconductor elements **10A** are electrically bonded to the conductive substrate **2** (a first conductive portion **2A** described below) via the second conductive bonding member **72**. When the first semiconductor elements **10A** are bonded to the first conductive portion **2A**, the element reverse surfaces **102** face the first conductive portion **2A**.

(44) As shown in FIGS. **8** and **17**, for example, the second semiconductor elements **10B** are mounted on the conductive substrate **2**. In the example shown in FIG. **8**, the second semiconductor elements **10B** are aligned in the y direction and spaced apart from each other. The second semiconductor elements **10B** are electrically bonded to the conductive substrate **2** (a second conductive portion **2B** described below) via the second conductive bonding member **72**. When the second semiconductor elements **10B** are bonded to the second conductive portion **2B**, the element reverse surfaces **102** face the second conductive portion **2B**. As is evident from FIG. **8**, the first semiconductor elements **10A** and the second semiconductor elements **10B** overlap with each other as viewed in the x direction, but they may not necessarily overlap with each other.

(45) Each of the semiconductor elements **10** (the first semiconductor elements **10A** and the second semiconductor elements **10B**) has a first obverse-surface electrode **11**, a second obverse-surface electrode **12**, and a reverse-surface electrode **15**. The configurations of the first obverse-surface electrode **11**, the second obverse-surface electrode **12**, and the reverse-surface electrode **15**, which are described below, are common to each of the semiconductor elements **10**. The first obverse-surface electrode **11** and the second obverse-surface electrode **12** are mounted on the element obverse surface **101**. The first obverse-surface electrode **11** and the second obverse-surface electrode **12** are insulated from each other by an insulating film (not illustrated). The reverse-surface electrode **15** is provided on the element reverse surface **102**.

(46) The first obverse-surface electrode **11** is a gate electrode, for example, to which a drive signal (e.g., gate voltage) for driving the semiconductor element **10** is inputted. In the semiconductor

element **10**, the second obverse-surface electrode **12** is a source electrode, for example, through which a source current flows. The reverse-surface electrode **15** is a drain electrode, for example, through which a drain current flows. The reverse-surface electrode **15** almost entirely covers the element reverse surface **102**. The reverse-surface electrode **15** may be formed by Ag plating.

(47) Each of the semiconductor elements **10** switches between a connected state and a disconnected state according to a drive signal (gate voltage), which is input to the first obverse-surface electrode **11** (gate electrode) via the switching function unit **Q1**. The operation of switching between the connected state and the disconnected state is referred to as a switching operation. In the connected state, a current flows from the reverse-surface electrode **15** (drain electrode) to the second obverse-surface electrode **12** (source electrode). In the disconnected state, the current does not flow. In other words, each of the semiconductor elements **10** performs a switching operation through the switching function unit **Q1**. The semiconductor module **A1** converts a first power supply voltage (DC voltage) inputted between the input terminal **41** and the two input terminals **42**, **43** to a second power supply voltage (AC voltage) by the switching function units **Q1** of the semiconductor elements **10**, for example, and outputs the second power supply voltage from the output terminals **44**. The input terminals **41** to **43** and the output terminals **44** are power supply terminals that handle power supply voltage. The input terminals **41** to **43** are first power supply terminals that receive the first source voltage. The output terminals **44** are second power supply terminals that output the second source voltage.

(48) Some (two in the example shown in FIG. **8**) of the semiconductor elements **10** each have a diode function unit **D1** (see FIG. **20**) in addition to the switching function unit **Q1**. In the semiconductor module **A1**, one of the first semiconductor elements **10A** (i.e., the first semiconductor element **10A** offset furthest in the y2 direction in FIG. **8**) and one of the second semiconductor elements **10B** (i.e., the second semiconductor element **10B** offset furthest in the y1 direction in FIG. **8**) each include a diode function unit **D1** in addition to the switching function unit **Q1**. The function and role of the diode function unit **D1** is not particularly limited but one example of the diode function unit **D1** is a temperature detection diode. Note that the diode **D2** in FIG. **20** is, for example, a parasitic diode component of the switching function unit **Q1**.

(49) As shown in FIG. **8**, each of the semiconductor elements **10** having the diode function units **D1** has a third obverse-surface electrode **13**, a fourth obverse-surface electrode **14**, and a fifth obverse-surface electrode **16**, in addition to the first obverse-surface electrode **11**, the second obverse-surface electrode **12**, and the reverse-surface electrode **15**. The configurations of the third obverse-surface electrode **13**, the fourth obverse-surface electrode **14**, and the fifth obverse-surface electrode **16**, which are described below, are common to each of the semiconductor elements **10** having the diode function units **D1**. The third obverse-surface electrode **13**, the fourth obverse-surface electrode **14**, and the fifth obverse-surface electrode **16** are formed on the element obverse surface **101**. In each of the semiconductor elements **10** having the diode function units **D1**, the third obverse-surface electrode **13** and the fourth obverse-surface electrode **14** are electrically connected to the diode function unit **D1**. The fifth obverse-surface electrode **16** is a source sense electrode, for example, through which a source current in the switching function unit **Q1** flows.

(50) As shown in FIG. **7**, each of the first semiconductor elements **10A** has a first side **191**, a second side **192**, a third side **193**, and a fourth side **194** in plan view. FIG. **7** illustrates the first semiconductor element **10A** arranged in the middle in the y direction among the first semiconductor elements **10A** aligned in the y direction, but each of the other first semiconductor elements **10A** also similarly has a first side **191**, a second side **192**, a third side **193**, and a fourth side **194**. The first side **191** and the second side **192** extend in the y direction. The first side **191** is an edge located in the x2 direction in plan view, and the second side **192** is an edge located in the x1 direction in plan view. The third side **193** and the fourth side **194** extend in the x direction. The third side **193** is an edge located in the y2 direction in plan view, and the fourth side **194** is an edge located in the y1 direction in plan view. Since each of the first semiconductor elements **10A** has a

rectangular shape in plan view, the four corners formed by the first side **191**, the second side **192**, the third side **193**, and the fourth side **194** are generally right-angled in plan view. As shown in FIG. 7, the four corners do not overlap with the conducting member **6** (first conducting members **61** and a second conducting member **62** described below) in plan view. The third side **193** and the fourth side **194** are longer than the first side **191** and the second side **192**.

(51) The conductive substrate **2** is also referred to as a lead frame. The conductive substrate **2** supports the semiconductor elements **10**. The conductive substrate **2** is bonded to the supporting substrate **3** via the first conductive bonding member **71**. The conductive substrate **2** has a rectangular shape in plan view, for example. The conductive substrate **2**, together with the conducting member **6**, forms the path of a main circuit current switched by the semiconductor elements **10**.

(52) The conductive substrate **2** includes a first conductive portion **2A** and a second conductive portion **2B**. The first conductive portion **2A** and the second conductive portion **2B** are plate-like members made of metal. The metal is copper (Cu) or a Cu alloy, for example. The first conductive portion **2A** and the second conductive portion **2B** form a conductive path to the semiconductor elements **10**, together with the input terminals **41** to **43** and the output terminals **44**. As shown in FIGS. 13 to 18, the first conductive portion **2A** and the second conductive portion **2B** are bonded to the supporting substrate **3** via the first conductive bonding member **71**. The first semiconductor elements **10A** are bonded to the first conductive portion **2A** via the second conductive bonding member **72**. The second semiconductor elements **10B** are bonded to the second conductive portion **2B** via the second conductive bonding member **72**. As shown in FIGS. 3, 8, 13, and 14, the first conductive portion **2A** and the second conductive portion **2B** are spaced apart from each other in the x direction. In the example shown in these figures, the first conductive portion **2A** is offset in the x2 direction relative to the second conductive portion **2B**. The first conductive portion **2A** and the second conductive portion **2B** each have a rectangular shape in plan view, for example. The first conductive portion **2A** and the second conductive portion **2B** overlap with each other as viewed in the x direction. The first conductive portion **2A** and the second conductive portion **2B** may each have a dimension of 15 mm to 25 mm (preferably about 20 mm) in the x direction, a dimension of 30 mm to 40 mm (preferably about 35 mm) in the y direction, and a dimension of 1.5 mm to 3.0 mm (preferably about 2.0 mm) in the z direction.

(53) The conductive substrate **2** has an obverse surface **201** and a reverse surface **202**. As shown in FIGS. 13, 14, and 16 to 18, the obverse surface **201** and the reverse surface **202** are spaced apart from each other in the z direction. The obverse surface **201** faces in the z2 direction, and the reverse surface **202** faces in the z1 direction. The obverse surface **201** is a combination of the upper surface of the first conductive portion **2A** and the upper surface of the second conductive portion **2B**. The reverse surface **202** is a combination of the lower surface of the first conductive portion **2A** and the lower surface of the second conductive portion **2B**. The reverse surface **202** faces the supporting substrate **3** and is bonded to the supporting substrate **3**. As shown in FIGS. 5, 8, and 13, the obverse surface **201** is formed with a plurality of recessed portions **201a**. The recessed portions **201a** are recessed from the obverse surface **201** in the z direction. The degree of recession (depth) of each recessed portion **201a** is greater than 0 μm and less than or equal to 100 μm , for example. The recessed portions **201a** are formed during molding described below, for example. The recessed portions **201a** include those formed in the obverse surface **201** of the first conductive portion **2A**, and those formed in the obverse surface **201** of the second conductive portion **2B**. Two recessed portions **201a** formed in the obverse surface **201** of the first conductive portion **2A** are spaced apart from each other in the y direction and overlap with each other as viewed in the y direction. Two recessed portions **201a** formed in the obverse surface **201** of the second conductive portion **2B** are spaced apart from each other in the y direction and overlap with each other as viewed in the y direction.

(54) The conductive substrate **2** (each of the first conductive portion **2A** and the second conductive

portion **2B**) includes a base member **21**, an obverse-surface bonding layer **22**, and a reverse-surface bonding layer **23** that are stacked on each other. The base member **21** is a plate-like member made of metal. The metal is Cu or a Cu alloy. The obverse-surface bonding layer **22** is formed on the upper surface of the base member **21**. The obverse-surface bonding layer **22** is the surface layer of the conductive substrate **2** in the z_2 direction. The upper surface of the obverse-surface bonding layer **22** corresponds to the obverse surface **201** of the conductive substrate **2**. The obverse-surface bonding layer **22** is a Ag plating layer, for example. The reverse-surface bonding layer **23** is formed on the lower surface of the base member **21**. The reverse-surface bonding layer **23** is the surface layer of the conductive substrate **2** in the z_1 direction. The lower surface of the reverse-surface bonding layer **23** corresponds to the reverse surface **202** of the conductive substrate **2**. The reverse-surface bonding layer **23** is a Ag plating layer, for example, as is the obverse-surface bonding layer **22**.

(55) The supporting substrate **3** supports the conductive substrate **2**. The supporting substrate **3** is a direct bonded copper (DBC) substrate, for example. The supporting substrate **3** includes an insulating layer **31**, a first metal layer **32**, a first bonding layer **321**, and a second metal layer **33**.

(56) The insulating layer **31** is made of a ceramic with excellent thermal conductivity, for example. The ceramic may be aluminum nitride (AlN). The insulating layer **31** is not limited to a ceramic, and may be an insulating resin sheet, for example. The insulating layer **31** has a rectangular shape in plan view, for example.

(57) The first metal layer **32** is formed on the upper surface (surface facing in the z_2 direction) of the insulating layer **31**. The material of the first metal layer **32** contains Cu, for example. The material may contain Al instead of Cu. The first metal layer **32** includes a first portion **32A** and a second portion **32B**. The first portion **32A** and the second portion **32B** are spaced apart from each other in the x direction. The first portion **32A** is offset in the x_2 direction relative to the second portion **32B**. The first portion **32A** is bonded to the first conductive portion **2A** and supports the first conductive portion **2A**. The second portion **32B** is bonded to the second conductive portion **2B** and supports the second conductive portion **2B**. The first portion **32A** and the second portion **32B** each have a rectangular shape in plan view, for example.

(58) The first bonding layer **321** is formed on the upper surface of the first metal layer **32** (each of the first portion **32A** and the second portion **32B**). The first bonding layer **321** is a Ag plating layer, for example. The first bonding layer **321** is provided to enhance the solid-phase diffusion bonding with the first conductive bonding member **71**.

(59) The second metal layer **33** is formed on the lower surface of the insulating layer **31** (surface facing in the z_1 direction). The second metal layer **33** is made of the same material as the first metal layer **32**. In the example shown in FIG. **10**, the lower surface (a bottom surface **302** described below) of the second metal layer **33** is exposed from the sealing resin **8**. The lower surface may not be exposed from the sealing resin **8**, and may be covered with the sealing resin **8**. The second metal layer **33** may overlap with the first portion **32A** and the second portion **32B** in plan view.

(60) As shown in FIGS. **13** to **18**, the supporting substrate **3** has a supporting surface **301** and a bottom surface **302**. The supporting surface **301** and the bottom surface **302** are spaced apart from each other in the z direction. The supporting surface **301** faces in the z_2 direction, and the bottom surface **302** faces in the z_1 direction. As shown in FIG. **10**, the bottom surface **302** is exposed from the sealing resin **8**. The supporting surface **301** is the upper surface of the first bonding layer **321**, i.e., a combination of the upper surface of the first portion **32A** and the upper surface of the second portion **32B**. The supporting surface **301** faces the conductive substrate **2**, and is bonded to the conductive substrate **2**. The bottom surface **302** is the lower surface of the second metal layer **33**. A heat dissipating member (e.g., heat sink) or the like, which is not illustrated in the figure, can be attached to the bottom surface **302**. The dimension of the supporting substrate **3** in the z direction (distance along the z direction from the supporting surface **301** to the bottom surface **302**) is 0.7 mm to 2.0 mm, for example.

(61) Each of the input terminals **41** to **43** and the output terminals **44** is a plate-like metal plate. The metal plate is made of Cu or a Cu alloy, for example. In the example shown in FIGS. **1** to **5**, **8**, and **10**, the semiconductor module **A1** includes the three input terminals **41** to **43** and the two output terminals **44**.

(62) Power supply voltage is applied to the three input terminals **41** to **43**. In the present embodiment, the input terminal **41** is a positive electrode (P terminal), and the two input terminals **42** and **43** are negative electrodes (N terminal). Alternatively, the input terminal **41** may be a negative terminal (N terminal), and the two input terminals **42** and **43** may be positive terminals (P terminals). In this case, the wiring in the package may be appropriately changed according to the change in the polarity of each terminal. Each of the three input terminals **41** to **43** and the two output terminals **44** includes a portion covered with the sealing resin **8** and a portion exposed from a resin side surface of the sealing resin **8**.

(63) As shown in FIG. **14**, the input terminal **41** is formed integrally with the first conductive portion **2A**. Unlike this configuration, the input terminal **41** may be separated from the first conductive portion **2A** and electrically bonded to the first conductive portion **2A**. As shown in FIG. **8**, for example, the input terminal **41** is offset in the x2 direction relative to the first semiconductor elements **10A** and the first conductive portion **2A** (conductive substrate **2**). The input terminal **41** is electrically connected to the first conductive portion **2A**, and is electrically connected to the reverse-surface electrodes **15** (drain electrodes) of the first semiconductor elements **10A** via the first conductive portion **2A**. The input terminal **41** is an example of a “first input terminal”.

(64) The input terminal **41** has an input-side bonding surface **411** and input-side side surfaces **412**. The input-side bonding surface **411** faces in the z2 direction and extends in the x2 direction. Each of the input-side side surfaces **412** is located at the periphery of the input-side bonding surface **411** as viewed in the z direction, and faces in a direction intersecting with the input-side bonding surface **411**. In the present embodiment, the input-side side surfaces **412** include a tip surface **413** and a pair of lateral surfaces **414**. The tip surface **413** is positioned at the end of the input terminal **41** in the x2 direction and faces in the x2 direction. The pair of lateral surfaces **414** are located at the respective ends of the input terminal **41** in the y direction, and face in the y1 direction and the y2 direction, respectively. Among the input-side side surfaces **412**, at least one of the tip surface **413** and the pair of lateral surfaces **414** has an input-side machining mark. The input-side machining mark is formed by the cutting process of a lead frame as described below.

(65) As shown in FIG. **8**, the two input terminals **42** and **43** are spaced apart from the first conductive portion **2A**. The two input terminals **42** and **43** are bonded to the second conducting member **62**. As shown in FIG. **8**, for example, the two input terminals **42** and **43** are offset in the x2 direction relative to the first semiconductor elements **10A** and the first conductive portion **2A** (conductive substrate **2**). The two input terminals **42** and **43** are electrically connected to the second conducting member **62**, and are electrically connected to the second obverse-surface electrodes **12** (source electrodes) of the second semiconductor elements **10B** via the second conducting member **62**. The input terminal **42** is an example of a “second input terminal”, and the input terminal **43** is an example of a “third input terminal”.

(66) The input terminal **42** has an input-side bonding surface **421** and input-side side surfaces **422**, and the input terminal **43** has an input-side bonding surface **431** and input-side side surfaces **432**. The input-side bonding surfaces **421** and **431** face in the z2 direction, and extend in the x2 direction. Each of the input-side side surfaces **422** is located at the periphery of the input-side bonding surface **421** as viewed in the z direction, and faces in a direction intersecting with the input-side bonding surface **421**. Each of the input-side side surfaces **432** is located at the periphery of the input-side bonding surface **431** as viewed in the z direction, and faces in a direction intersecting with the input-side bonding surface **431**. In the present embodiment, the input-side side surfaces **422** include a tip surface **423** and a pair of lateral surfaces **424**. The tip surface **423** is positioned at the end of the input terminal **42** in the x2 direction and faces in the x2 direction. The

pair of lateral surfaces **424** are located at the respective ends of the input terminal **42** in the y direction, and face in the y1 direction and the y2 direction, respectively. Among the input-side side surfaces **422**, at least one of the tip surface **423** and the pair of lateral surfaces **424** has an input-side machining mark. The input-side machining mark is formed by the cutting process of a lead frame as described below. The input-side side surfaces **432** include a tip surface **433** and a pair of lateral surfaces **434**. The tip surface **433** is positioned at the end of the input terminal **43** in the x2 direction and faces in the x2 direction. The pair of lateral surfaces **434** are located at the respective ends of the input terminal **43** in the y direction, and face in the y1 direction and the y2 direction, respectively. Among the input-side side surfaces **432**, at least one of the tip surface **433** and the pair of lateral surfaces **434** has an input-side machining mark. The input-side machining mark is formed by the cutting process of a lead frame as described below.

(67) As shown in FIGS. **1** to **5**, **8**, and **10**, for example, the three input terminals **41** to **43** of the semiconductor module **A1** protrude from the sealing resin **8** in the x2 direction. The three input terminals **41** to **43** are spaced apart from each other. The two input terminals **42** and **43** are located opposite from each other with the input terminal **41** therebetween in the y direction. The input terminal **42** is offset in the y2 direction relative to the input terminal **41**, and the input terminal **43** is offset in the y1 direction relative to the input terminal **41**. The three input terminals **41** to **43** overlap with each other as viewed in the y direction.

(68) As is evident from FIGS. **8** and **14**, the two output terminals **44** are integrally formed with the second conductive portion **2B**. Unlike this configuration, the output terminals **44** may be separated from the second conductive portion **2B** and electrically bonded to the second conductive portion **2B**. As shown in FIG. **8**, for example, the two output terminals **44** are offset in the x1 direction relative to the second semiconductor elements **10B** and the second conductive portion **2B**

(conductive substrate **2**). The output terminals **44** are electrically connected to the second conductive portion **2B**, and are electrically connected to the reverse-surface electrodes **15** (drain electrodes) of the second semiconductor elements **10B** via the second conductive portion **2B**. The two output terminals **44** are examples of a “first output terminal” and a “second output terminal”.

(69) Each of the output terminals **44** has an output-side bonding surface **441** and output-side side surfaces **442**. The output-side bonding surface **441** faces in the z2 direction and extends in the x1 direction. Each of the output-side side surfaces **442** is located at the periphery of the output-side bonding surface **441** as viewed in the z direction, and faces in a direction intersecting with the output-side bonding surface **441**. In the present embodiment, the output-side side surfaces **442** include a tip surface **443** and a pair of lateral surfaces **444**. The tip surface **443** is positioned at the end of the output terminal **44** in the x1 direction and faces in the x1 direction. The pair of lateral surfaces **444** are located at the respective ends of the output terminal **44** in the y direction, and face in the y1 direction and the y2 direction, respectively. Among the output-side side surfaces **442**, at least one of the tip surface **443** and the pair of lateral surfaces **444** has an output-side machining mark. The output-side machining mark is formed by the cutting process of a lead frame as described below. The number of output terminals **44** is not limited to two, and may be one or no less than three. When the number of output terminals **44** is one, it is desirable that the output terminal **44** be connected to the middle section of the second conductive portion **2B** in the y direction.

(70) The control terminals **45** are pin-like terminals for controlling the semiconductor elements **10**. The control terminals **45** include a plurality of first control terminals **46A** to **46E** and a plurality of second control terminals **47A** to **47D**. The first control terminals **46A** to **46E** are used to control the first semiconductor elements **10A**. The second control terminals **47A** to **47D** are used to control the second semiconductor elements **10B**.

(71) The first control terminals **46A** to **46E** are arranged at intervals in the y direction. As shown in FIGS. **8** and **14**, for example, the first control terminals **46A** to **46E** are supported by the first conductive portion **2A** via the control terminal support **5** (a first supporting portion **5A** described

below). As shown in FIGS. 5 and 8, the first control terminals **46A** to **46E** are located between the first semiconductor elements **10A** and the three input terminals **41** to **43** in the x direction.

(72) The first control terminal **46A** is a terminal (gate terminal) used to input a drive signal for the first semiconductor elements **10A**. The first control terminal **46A** receives the drive signal for driving the first semiconductor elements **10A** (e.g., it receives application of gate voltage).

(73) The first control terminal **46B** is a terminal (source sense terminal) used to detect a source signal for the first semiconductor elements **10A**. Voltage (corresponding to a source current) applied to the second obverse-surface electrodes **12** (source electrodes) of the first semiconductor elements **10A** is detected from the first control terminal **46B**.

(74) The first control terminals **46C** and **46D** are terminals that are electrically connected to the diode function unit **D1**. The first control terminal **46C** is electrically connected to the third obverse-surface electrode **13** of the first semiconductor element **10A** having the diode function unit **D1**, and the first control terminal **46D** is electrically connected to the fourth obverse-surface electrode **14** of the first semiconductor element **10A** having the diode function unit **D1**.

(75) The first control terminal **46E** is a terminal (drain sense terminal) used to detect a drain signal for the first semiconductor elements **10A**. Voltage (corresponding to a drain current) applied to the reverse-surface electrodes **15** (drain electrodes) of the first semiconductor elements **10A** is detected from the first control terminal **46E**.

(76) The second control terminals **47A** to **47D** are arranged at intervals in the y direction. As shown in FIGS. 5 and 18, for example, the second control terminals **47A** to **47D** are supported by the second conductive portion **2B** via the control terminal support **5** (a second supporting portion **5B** described below). As shown in FIGS. 5 and 8, the second control terminals **47A** to **47D** are located between the second semiconductor elements **10B** and the two output terminals **44** in the x direction.

(77) Each of the control terminals **45** (first control terminals **46A** to **46E** and the second control terminals **47A** to **47D**) includes a holder **451** and a metal pin **452**.

(78) The holder **451** is made of a conductive material. As shown in FIG. 15, the holder **451** is bonded to the control terminal support **5** (a first metal layer **52** described below) via a conductive bonding member **459**. The holder **451** includes a tubular portion, an upper-end flange portion, and a lower-end flange portion. The upper-end flange portion is joined to the top of the tubular portion, and the lower-end flange portion is joined to the bottom of the tubular portion. The metal pin **452** is inserted through at least the upper-end flange portion and the tubular portion of the holder **451**. The upper surface of the upper-end flange portion is exposed from the sealing resin **8** (a second protruding portion **852** described below), and is covered with the resin member **87**.

(79) The metal pin **452** is a rod-like member extending in the z direction. The metal pin **452** is supported by being pressed into the holder **451**. The metal pin **452** is electrically connected to the control terminal support **5** (a first metal layer **52** described below) at least via the holder **451**. As shown in the example in FIG. 15, when the lower end (end in the z1 direction) of the metal pin **452** is in contact with the conductive bonding member **459** within the insertion hole of the holder **451**, the metal pin **452** is electrically connected to the control terminal support **5** via the conductive bonding member **459**.

(80) The control terminal support **5** supports the control terminals **45**. The control terminal support **5** is provided between the obverse surface **201** (conductive substrate **2**) and the control terminals **45**.

(81) The control terminal support **5** includes a first supporting portion **5A** and a second supporting portion **5B**. The first supporting portion **5A** is arranged on the first conductive portion **2A** of the conductive substrate **2**, and supports the first control terminals **46A** to **46E** among the control terminals **45**. As shown in FIG. 15, the first supporting portion **5A** is bonded to the first conductive portion **2A** via a bonding member **59**. The bonding member **59** may be conductive or insulative. For example, the bonding member **59** may be solder. The second supporting portion **5B** is arranged

on the second conductive portion 2B of the conductive substrate 2, and supports the second control terminals 47A to 47D among the control terminals 45. The second supporting portion 5B is bonded to the second conductive portion 2B via the bonding member 59.

(82) The control terminal support 5 (the first supporting portion 5A and the second supporting portion 5B) may be a DBC substrate, for example. Each of the supporting portions of the control terminal support 5 includes an insulating layer 51, a first metal layer 52, and a second metal layer 53 that are stacked on each other.

(83) The insulating layer 51 is made of a ceramic, for example. The insulating layer 51 has a rectangular shape in plan view, for example.

(84) As shown in FIG. 15, for example, the first metal layer 52 is formed on the upper surface of the insulating layer 51. The control terminals 45 are erected on the first metal layer 52. The first metal layer 52 is made of Cu or a Cu alloy, for example. As shown in FIG. 8, the first metal layer 52 includes a first portion 521, a second portion 522, a third portion 523, a fourth portion 524, and a fifth portion 525. The first portion 521, the second portion 522, the third portion 523, the fourth portion 524, and the fifth portion 525 are spaced apart and insulated from each other.

(85) A plurality of wires 731 are bonded to the first portion 521, so that the first portion 521 is electrically connected to the first obverse-surface electrodes 11 (gate electrodes) of the semiconductor elements 10 via the wires 731. As shown in FIG. 8, the first control terminal 46A is bonded to the first portion 521 of the first supporting portion 5A, and the second control terminal 47A is bonded to the first portion 521 of the second supporting portion 5B.

(86) A plurality of wires 732 are bonded to the second portion 522, so that the second portion 522 is electrically connected to the second obverse-surface electrodes 12 (source electrodes) of the semiconductor elements 10 via the wires 732. As shown in FIG. 8, the first control terminal 46B is bonded to the second portion 522 of the first supporting portion 5A, and the second control terminal 47B is bonded to the second portion 522 of the second supporting portion 5B.

(87) A wire 733 is bonded to the third portion 523, so that the third portion 523 is electrically connected to the third obverse-surface electrode 13 of the semiconductor element 10 having the diode function unit D1 via the wire 733. As shown in FIG. 8, the first control terminal 46C is bonded to the third portion 523 of the first supporting portion 5A, and the second control terminal 47C is bonded to the third portion 523 of the second supporting portion 5B.

(88) A wire 734 is bonded to the fourth portion 524, so that the fourth portion 524 is electrically connected to the fourth obverse-surface electrode 14 of the semiconductor element 10 having the diode function unit D1 via the wire 734. As shown in FIG. 8, the first control terminal 46D is bonded to the fourth portion 524 of the first supporting portion 5A, and the second control terminal 47D is bonded to the fourth portion 524 of the second supporting portion 5B.

(89) A wire 735 is bonded to the fifth portion 525 of the first supporting portion 5A, and the fifth portion 525 is electrically connected to the first conductive portion 2A. The fifth portion 525 of the second supporting portion 5B is not electrically connected to other components. As shown in FIG. 8, the first control terminal 46E is bonded to the fifth portion 525 of the first supporting portion 5A.

(90) As shown in FIG. 15, for example, the second metal layer 53 is formed on the lower surface of the insulating layer 51. As shown in FIG. 15, the second metal layer 53 of the first supporting portion 5A is bonded to the first conductive portion 2A via the bonding member 59. The second metal layer 53 of the second supporting portion 5B is bonded to the second conductive portion 2B via the bonding member 59.

(91) The conducting member 6, together with the conductive substrate 2, forms the path of a main circuit current switched by the semiconductor elements 10. The conducting member 6 is separated from the obverse surface 201 (conductive substrate 2) in the z2 direction, and overlaps with the obverse surface 201 in plan view. In the present embodiment, the conducting member 6 is made of a metal plate-like member. The metal is Cu or a Cu alloy, for example. Specifically, the conducting member 6 is a metal plate-like member that is bent. Alternatively, the conducting member 6 may be

formed with a metal foil member. In the present embodiment, the conducting member **6** includes a plurality of first conducting members **61** and a second conducting member **62**. The main circuit current includes a first main circuit current and a second main circuit current. The first main circuit current flows through a path between the input terminal **41** and the output terminals **44**. The second main circuit current flows through a path between the output terminals **44** and the input terminals **42, 43**.

(92) The first conducting members **61** are connected to the second obverse-surface electrodes **12** (source electrodes) of the first semiconductor elements **10A** and the second conductive portion **2B**, so that the second obverse-surface electrodes **12** of the first semiconductor elements **10A** are electrically connected to the second conductive portion **2B**. The first conducting members **61** and the second obverse-surface electrodes **12** (see FIG. **8**) of the first semiconductor elements **10A**, as well as the first conducting members **61** and the second conductive portion **2B**, are bonded to each other via the conductive bonding member **69**. The conductive bonding member **69** may be made of solder, a metal paste material, or a sintered metal. As shown in FIG. **8**, each of the first conducting members **61** has a band shape extending along the x direction in plan view.

(93) In the present embodiment, each of the first conducting members **61** has a rectangular portion connecting the first semiconductor element **10A** and the second conductive portion **2B**, and the rectangular portion is formed with an opening **61h**, as shown in FIG. **6**, for example. The opening **61h** may be a through-hole that penetrates through in the z direction, and is preferably formed in the center of the rectangular portion in plan view. The opening **61h** is formed so that when a flowable resin material is injected to form a sealing resin, the resin material can easily flow between the upper side (in the z2 direction) and the lower side (in the z1 direction) near the first conducting member **61**. In plan view, the opening **61h** may have a perfectly circular shape, or may have another shape such as an oval shape or a rectangular shape. Each of the first conducting members **61** is not limited to having the configuration described above, and may not be formed with an opening **61h**.

(94) In the present embodiment, the number of first conducting members **61** is three so as to correspond to the number of first semiconductor elements **10A**. Alternatively, a single first conducting member **61** common to the first semiconductor elements **10A** may be used, without depending on the number of first semiconductor elements **10A**.

(95) The second conducting member **62** electrically connects the second obverse-surface electrodes **12** of the second semiconductor elements **10B** to the input terminals **42** and **43**. The second conducting member **62** may have a maximum dimension of 25 mm to 40 mm (preferably about 32 mm) in the x direction, and a maximum dimension of 30 mm to 45 mm (preferably about 38 mm) in the y direction. As shown in FIG. **6**, the second conducting member **62** includes a first wiring portion **621**, a second wiring portion **622**, a third wiring portion **623**, and a fourth wiring portion **624**.

(96) The first wiring portion **621** is connected to the input terminal **42**. The first wiring portion **621** and the input terminal **42** are bonded with the conductive bonding member **69**. The first wiring portion **621** has a band shape extending in the x direction in plan view.

(97) The second wiring portion **622** is connected to the input terminal **43**. The second wiring portion **622** and the input terminal **43** are bonded with the conductive bonding member **69**. The second wiring portion **622** has a band shape extending in the x direction in plan view. The first wiring portion **621** and the second wiring portion **622** are spaced apart from each other in the y direction and arranged substantially in parallel to each other. The second wiring portion **622** is offset in the y1 direction relative to the first wiring portion **621**.

(98) The third wiring portion **623** is joined to the first wiring portion **621** and the second wiring portion **622**. The third wiring portion **623** has a band shape extending in the y direction in plan view. As is evident from FIG. **6**, the third wiring portion **623** overlaps with the second semiconductor elements **10B** in plan view. As shown in FIG. **17**, the third wiring portion **623** is

connected to the second semiconductor elements **10B**. The third wiring portion **623** has a plurality of recessed areas **623a**. As shown in FIG. **17**, the recessed areas **623a** are recessed in the **z1** direction relative to the other areas of the third wiring portion **623**. The recessed areas **623a** of the third wiring portion **623** are bonded to the second semiconductor elements **10B**. The recessed areas **623a** of the third wiring portion **623** and the second obverse-surface electrodes **12** (see FIG. **8**) of the second semiconductor elements **10B** are bonded with the conductive bonding member **69**.

(99) The fourth wiring portion **624** is joined to the first wiring portion **621** and the second wiring portion **622**. The fourth wiring portion **624** is also connected to the third wiring portion **623**. The fourth wiring portion **624** is offset in the **x2** direction relative to the third wiring portion **623**. As is evident from FIG. **6**, the fourth wiring portion **624** overlaps with the first semiconductor elements **10A** in plan view. The fourth wiring portion **624** includes a first band portion **625** and a plurality of second band portions **626**.

(100) The first band portion **625** is a part of the fourth wiring portion **624** that has a band shape in plan view, and is spaced apart from the third wiring portion **623** in the **x** direction. The first band portion **625** is joined to the first wiring portion **621** and the second wiring portion **622**. The first band portion **625** overlaps with the first semiconductor elements **10A** in plan view. The first band portion **625** has a plurality of protruding areas **625a**. As shown in FIG. **16**, the protruding areas **625a** protrude in the **z2** direction relative to the other areas of the first band portion **625**. As shown in FIG. **6**, the protruding areas **625a** overlap with the first semiconductor elements **10A** in plan view. Since the first band portion **625** has the protruding areas **625a**, areas for bonding the first conducting members **61** are provided on the first semiconductor elements **10A**, as shown in FIG. **16**. This prevents the first band portion **625** from being in contact with the first conducting members **61**.

(101) Each of the second band portions **626** is connected to the first band portion **625** and the third wiring portion **623**. Each of the second band portions **626** has a band shape extending in the **x** direction in plan view. The second band portions **626** are spaced apart from each other in the **y** direction and arranged substantially in parallel to each other. In plan view, one end of each band portion **626** is connected to a part of the first band portion **625**, which is located between two first semiconductor elements **10A** adjacent in the **y** direction, and the other end of each band portion **626** is connected to a part of the third wiring portion **623**, which is located between two second semiconductor elements **10B** adjacent in the **y** direction.

(102) The first band portion **625** has a first edge **627** and a second edge **628**. As shown in FIG. **7**, the first edge **627** is offset in the **x1** direction relative to the first side **191** in plan view, and extends at least from the third side **193** to the fourth side **194** in the **y** direction. As such, two corners **171** and **172** of each first semiconductor element **10A** in the **x2** direction do not overlap with the second conducting member **62** in plan view. The two corners are the corner **171** formed by the first side **191** and the third side **193**, and the corner **172** formed by the first side **191** and the fourth side **194**. Accordingly, in each of the first semiconductor elements **10A**, parts of the two sides flanking the corners **171** and **172** are visible in plan view (specifically, when viewed as shown in FIG. **7**; the same applies hereinafter). As shown in FIG. **7**, the second edge **628** is offset in the **x2** direction relative to the second side **192** in plan view, and extends at least from the third side **193** to the fourth side **194** in the **y** direction. As such, two corners **173** and **174** of each first semiconductor element **10A** in the **x1** direction do not overlap with the second conducting member **62** in plan view. The two corners are the corner **173** formed by the second side **192** and the third side **193**, and the corner **174** formed by the second side **192** and the fourth side **194**. Accordingly, in each of the first semiconductor elements **10A**, parts of the two sides flanking the corners **173** and **174** are visible in plan view.

(103) Regarding the corners **171**, **172**, **173**, and **174**, it is sufficient for each of the visible portions of the two sides flanking the corners **171**, **172**, **173**, and **174** to have a length greater than $0\ \mu\text{m}$ and no greater than $200\ \mu\text{m}$ in plan view. Furthermore, it is preferable that in plan view, the length of

each of the visible portions of the two sides flanking the corners **171**, **172**, **173**, and **174** be no less than 5 μm and no greater than 150 μm . When the length of each of the visible portions of the two sides flanking the corners **171**, **172**, **173**, and **174** is no less than 2 μm , it is possible to detect the corners of each of the first semiconductor elements **10A**. When the length of each of the visible portions of the two sides is no less than 5 μm , it is possible to reliably detect the corners of each of the first semiconductor elements **10A**. When the length of each of the visible portions of the two sides is greater than 200 μm , the bonding areas between the first conducting members **61** and the first semiconductor elements **10A** become smaller than necessary, which is not desirable. It is preferable that the upper limit of the length of each of the visible portions of the two sides be no greater than 150 μm , because the bonding area between the first conducting members **61** and the first semiconductor elements **10A** is prevented from being too small.

(104) As shown in FIG. 6, the conducting member **6** (first conducting members **61** and the second conducting member **62**) has first portions **601**. The first portions **601** are areas that overlap with the semiconductor elements **10** (the first semiconductor elements **10A** and the second semiconductor elements **10B**) in plan view. At the second conducting member **62**, parts of the fourth wiring portion **624** (areas overlapping with the first semiconductor elements **10A** in plan view) and parts of the third wiring portion **623** (areas overlapping with the second semiconductor elements **10B** in plan view) constitute the first portions **601**.

(105) As shown in FIGS. 6 and 8, the obverse-surface electrodes **11**, **13**, **14**, and **16** of one of the first semiconductor elements **10A** (the first semiconductor element **10A** having the diode function unit **D1**) are aligned along the y direction at the end of the first semiconductor element **10A** in the x2 direction. In plan view, the first conducting members **61** and the second conducting member **62** do not overlap with the obverse-surface electrodes **11**, **13**, **14**, and **16** of the first semiconductor element **10A** or with the corners **171** and **172** thereof in the x2 direction. Furthermore, in plan view, the first conducting members **61** and the second conducting member **62** do not overlap with at least one of the corners **173** and **174** of the first semiconductor element **10A** in the x1 direction (opposite from the side where the obverse-surface electrodes are arranged). As such, at least three corners among the four corners **171**, **172**, **173**, and **174** of the semiconductor element **10A** are visible in plan view. This makes it possible to inspect whether the first semiconductor elements **10A** are correctly mounted by automatic visual inspection when the first semiconductor elements **10A**, the conducting members **61**, and the second conducting member **62** are mounted on the conductive substrate **2**. In plan view, the four corners **171**, **172**, **173**, and **174** of each of the first semiconductor elements **10A** may all be visible. Note that the obverse-surface electrodes **11**, **13**, **14**, and **16** of the first semiconductor element **10A** are examples of “obverse-surface electrodes on one side”.

(106) As shown in FIG. 6, each of the second semiconductor elements **10B** has a rectangular shape in plan view, similarly to the first semiconductor elements **10A**, and has four corners **181**, **182**, **183**, and **184** corresponding to the four corners **171**, **172**, **173**, and **174** of each first semiconductor element **10A**. The above-described relationship in plan view between the four corners **171**, **172**, **173**, and **174** of each first semiconductor element **10A** and the first and second conducting members **61**, **62** also holds for the relationship in plan view between the four corners **181**, **182**, **183**, and **184** of each second semiconductor element **10B** and the second conducting member **62**.

(107) As shown in FIG. 5, the second conducting member **62** includes first portions **62A** and second portions **62B**. The first portions **62A** overlap with the obverse surface **201** of the conductive substrate **2** (the obverse surface **201** of either the first conductive portion **2A** or the second conductive portion **2B**) in plan view, and do not overlap with any of the semiconductor elements **10** in plan view. The second portions **62B** overlap with the obverse surface **201** in plan view, and overlap with the semiconductor elements **10** in plan view. In FIG. 5, the first portions **62A** are shown with hatching that diagonally rising to the right, and the second portions **62B** are shown with hatching that diagonally falling to the right. The first portions **62A** have openings **63**. As shown in FIGS. 5 and 13, for example, the openings **63** are partially cut away portions in plan view.

In the present embodiment, the openings **63** are provided at positions that overlap with the obverse surface **201** of the first conductive portion **2A** (conductive substrate **2**) in plan view, and that do not overlap with the semiconductor elements **10** in plan view. The openings **63** are through-holes that penetrate through in the z direction, for example. The openings **63** include one formed in the first wiring portion **621** and one formed in the second wiring portion **622**. The openings **63** are provided in the vicinity of at least two of the four corners of the conductive substrate **2** in plan view. For example, one of the openings **63** is provided at an area of the first wiring portion **621** in the x2 direction, and the other at an area of the second wiring portion **622** in the x2 direction. Note that the planar shape of each opening **63** is not limited. For example, the openings **63** may be holes as described in the present embodiment or notches unlike the present embodiment. The openings **63** may be formed by electroforming, for example. In this case, the second conducting member **62** has openings **63** in areas not electrodeposited with a metal, instead of the openings **63** formed by removing portions of the second conducting member **62**.

(108) The second conducting member **62** is formed with openings **625h** in rectangular portions that overlap with the first semiconductor elements **10A** in plan view. In the present embodiment, it is preferable that the openings **625h** be formed to overlap with the centers of the first semiconductor elements **10A** in plan view. The openings **625h** are through-holes (see FIG. 6) formed in the protruding areas **625a** of the first band portion **625** (fourth wiring portion **624**), for example. The openings **625h** are used, when the first conducting members **61** and the first semiconductor elements **10A** are bonded, to optically check the bonding state from above.

(109) The second conducting member **62** is formed with openings **623h** in rectangular portions that overlap with the second semiconductor elements **10B** in plan view. In the present embodiment, it is preferable that the openings **623h** be formed to overlap with the centers of the second semiconductor elements **10B** in plan view. The openings **623h** are through-holes formed in the recessed areas **623a** of the third wiring portion **623**, for example. The openings **623h** are used when the second conducting member **62** is positioned relative to the conductive substrate **2**. In plan view, each of the two types of openings **623h** and **625h** may have a perfectly circular shape, or may have another shape such as an oval shape or a rectangular shape.

(110) The second conducting member **62** is not limited to having the configuration described above, and may not include the fourth wiring portion **624**. However, the second conducting member **62** is preferably provided with the fourth wiring portion **624** in order to reduce the inductance value due to the current flowing through the second conducting member **62**.

(111) The first conductive bonding member **71** is provided between the conductive substrate **2** and the supporting substrate **3** to electrically bond the conductive substrate **2** and the supporting substrate **3**. The first conductive bonding member **71** includes a conductive bonding portion that electrically bonds the first conductive portion **2A** to the first portion **32A**, and a conductive bonding portion that electrically bonds the second conductive portion **2B** to the second portion **32B**. As shown in FIG. 15, the first conductive bonding member **71** includes a first base layer **711**, a first layer **712**, and a second layer **713** that are stacked on each other.

(112) As shown in FIG. 15, it is most preferable that a side surface of the first conductive bonding member **71** and a side surface of the first metal layer **32**, which is the top layer of the supporting substrate **3**, be flush with each other. It is preferable that in plan view, the side surface of the first metal layer **32** is positioned slightly more inward than the side surface of the first conductive bonding member **71**. That is, in plan view, bonding is performed such that the side surface of the first metal layer **32** does not extend outward from the side surface of the first conductive bonding member **71**. If the side surface of the first metal layer **32** extends more outward than the side surface of the first conductive bonding member **71** in plan view, the creepage distance between the first metal layer **32** and the second metal layer **33** becomes undesirably small. In plan view, the side surface of the first metal layer **32** is positioned more outward than a side surface of the base member **21** in the conductive substrate **2**.

(113) The first base layer **711** is made of a metal, such as Al or an Al alloy. The first base layer **711** is made of a sheet material. The Young's modulus of aluminum (Al), which is the material of the first base layer **711**, is 70.3 GPa.

(114) The first layer **712** is formed on the upper surface of the first base layer **711**. The first layer **712** is provided between the first base layer **711** and the conductive substrate **2** (each of the first conductive portion **2A** and the second conductive portion **2B**). The first layer **712** is a Ag plating layer, for example. The first layer **712** is bonded to the respective reverse-surface bonding layers **23** of the first conductive portion **2A** and the second conductive portion **2B** by the solid-phase diffusion of metal, for example. In other words, the first layer **712** and the reverse-surface bonding layers **23** of the first conductive portion **2A** and the second conductive portion **2B** are bonded by solid-phase diffusion. As a result, the first layer **712** and the reverse-surface bonding layers **23** are bonded in direct contact with each other at the bonding interface. In the present disclosure, "A and B are bonded by solid-phase diffusion" means that as a result of solid-phase diffusion bonding, A and B are fixed to each other in direct contact at the bonding interface, where A and B constitute a solid-phase diffusion layer. When solid-phase diffusion bonding is performed under an ideal condition, the bonding interface may not exist clearly due to the diffusion of metal elements. On the other hand, when an inclusion such as an oxidation film is formed on the surface layers of A and B, or when there is a gap between A and B, the inclusion or the gap may exist at the bonding interface.

(115) The second layer **713** is formed on the lower surface of the first base layer **711**. The second layer **713** is provided between the first base layer **711** and the supporting substrate **3** (each of the first portion **32A** and the second portion **32B**). The second layer **713** is a Ag plating layer, for example. The second layer **713** is bonded to the first bonding layer **321** formed on each of the first portion **32A** and the second portion **32B** by solid-phase diffusion of metal. In other words, the second layer **713** and the first bonding layer **321** are bonded by solid-phase diffusion in direct contact with each other at the bonding interface. The Young's modulus of silver (Ag), which is the material of the first layer **712** and the second layer **713**, is 82.7 GPa.

(116) Since the first base layer **711**, the first layer **712**, and the second layer **713** in the first conductive bonding member **71** are made of the materials described above, the Young's modulus of the first base layer **711** is smaller than the Young's modulus of each of the first layer **712** and the second layer **713**. The thickness (dimension in the z direction) of the first base layer **711** is greater than the thickness of each of the first layer **712** and the second layer **713**.

(117) In the first conductive bonding member **71**, an end surface of the first base layer **711**, which is made of Al or an Al alloy, is not plated with Ag, so that the end surface of the first base layer **711** is exposed. Note that the end surface of the first base layer **711** may be plated with Ag. In view of reducing the manufacturing cost of the first conductive bonding member **71**, it is preferable to fabricate the first conductive bonding member **71** by forming Ag plating on both surfaces of a large sheet material and then cutting the Ag-plated sheet material. In this regard, it is preferable that the end surface of the first base layer **711** not be plated with Ag.

(118) The second conductive bonding member **72** is provided between the conductive substrate **2** and the semiconductor elements **10** to electrically bond the conductive substrate **2** and the semiconductor elements **10**. The second conductive bonding member **72** includes a conductive bonding portion that electrically bonds the first semiconductor elements **10A** to the first conductive portion **2A**, and a conductive bonding portion that electrically bonds the second semiconductor elements **10B** to the second conductive portion **2B**. As shown in FIG. 15, the second conductive bonding member **72** includes a second base layer **721**, a third layer **722**, and a fourth layer **723** that are stacked on each other.

(119) The second base layer **721** is made of a metal, such as Al or an Al alloy. The second base layer **721** is made of a sheet material.

(120) The third layer **722** is formed on the upper surface of the second base layer **721**. The third

layer 722 is provided between the second base layer 721 and the semiconductor elements 10. The third layer 722 is a Ag plating layer, for example. The third layer 722 is bonded to the reverse-surface electrodes 15 of the semiconductor elements 10 by the solid-phase diffusion of metal, for example. In other words, the third layer 722 and the reverse-surface electrodes 15 are bonded by solid-phase diffusion in direct contact with each other at the bonding interface.

(121) The fourth layer 723 is formed on the lower surface of the second base layer 721. The fourth layer 723 is provided between the second base layer 721 and the conductive substrate 2 (each of the first conductive portion 2A and the second conductive portion 2B). The fourth layer 723 is a Ag plating layer, for example. The fourth layer 723 is bonded to the respective obverse-surface bonding layers 22 of the first conductive portion 2A and the second conductive portion 2B by the solid-phase diffusion of metal, for example. In other words, the fourth layer 723 and the obverse-surface bonding layers 22 are bonded by solid-phase diffusion in direct contact with each other at the bonding interface.

(122) Since the second base layer 721, the third layer 722, and the fourth layer 723 in the second conductive bonding member 72 are made of the materials described above, the Young's modulus of the second base layer 721 is smaller than the Young's modulus of each of the third layer 722 and the fourth layer 723. The thickness (dimension in the z direction) of the second base layer 721 is greater than the thickness of each of the third layer 722 and the fourth layer 723.

(123) In the second conductive bonding member 72, an end surface of the second base layer 721, which is made of Al or an Al alloy, is not plated with Ag, so that the end surface of the second base layer 721 is exposed. Note that the end surface of the second base layer 721 may be plated with Ag. In view of reducing the manufacturing cost of the second conductive bonding member 72, it is preferable to fabricate the second conductive bonding member 72 by forming Ag plating on both surfaces of a large sheet material and then cutting the Ag-plated sheet material. In this regard, it is preferable that the end surface of the second base layer 721 not be plated with Ag.

(124) Each of the wires 731 to 735 electrically connects two members that are separated from each other. The wires 731 to 735 are bonding wires, for example. The material of each of the wires 731 to 735 contains one of gold (Au), Al, or Cu, for example.

(125) As shown in FIG. 8, each of the wires 731 is bonded to and electrically connects the first obverse-surface electrode 11 (gate electrode) of a semiconductor element 10 and a first portion 521 (first metal layer 52) of the control terminal support 5. As shown in FIG. 8, the plurality of wires 731 include a plurality of first wires 731a and a plurality of second wires 731b. Each of the first wires 731a is connected to the first obverse-surface electrode 11 (gate electrode) of one of the first semiconductor elements 10A and the first portion 521 (first metal layer 52) of the first supporting portion 5A. As a result, the first control terminal 46A is electrically connected to the first obverse-surface electrodes 11 (gate electrodes) of the first semiconductor elements 10A via the first wires 731a. Each of the second wires 731b is connected to the first obverse-surface electrode 11 (gate electrode) of one of the second semiconductor elements 10B and the first portion 521 (first metal layer 52) of the second supporting portion 5B. As a result, the second control terminal 47A is electrically connected to the first obverse-surface electrodes 11 (gate electrodes) of the second semiconductor elements 10B via the second wires 731b.

(126) As shown in FIG. 8, each of the wires 732 is bonded to and electrically connects the second obverse-surface electrode 12 (source electrode) of a semiconductor element 10 and a second portion 522 (first metal layer 52) of the control terminal support 5. As for each of the semiconductor elements 10 having the diode function units D1, the wire 732 is bonded to the fifth obverse-surface electrode 16 (source sense electrode) instead of the second obverse-surface electrode 12 (source electrode).

(127) As shown in FIG. 8, each of the wires 733 is bonded to and electrically connects the third obverse-surface electrode 13 of one of the semiconductor elements 10 having the diode function units D1 and a third portion 523 (first metal layer 52) of the control terminal support 5.

(128) As shown in FIG. 8, each of the wires **734** is bonded to and electrically connects the fourth obverse-surface electrode **14** of one of the semiconductor elements **10** having the diode function units **D1** and a fourth portion **524** (first metal layer **52**) of the control terminal support **5**.

(129) As shown in FIG. 8, the wire **735** is bonded to and electrically connects the obverse surface **201** of the first conductive portion **2A** (conductive substrate **2**) and the fifth portion **525** (first metal layer **52**) of the first supporting portion **5A** (control terminal support **5**).

(130) The sealing resin **8** covers the semiconductor elements **10**, the conductive substrate **2**, the supporting substrate **3** (except the bottom surface **302**), parts of the input terminals **41** to **43**, parts of the output terminals **44**, parts of the control terminals **45**, the control terminal support **5**, the conducting member **6**, and the wires **731** to **735**. The sealing resin **8** is made of a black epoxy resin, for example. The sealing resin **8** may be formed by molding described below. The sealing resin **8** may have a dimension of about 35 mm to 60 mm in the x direction, a dimension of about 35 mm to 50 mm in the y direction, and a dimension of about 4 mm to 15 mm in the z direction. Each of these dimensions is the size of the largest portion along one of the directions. The sealing resin **8** has a resin obverse surface **81**, a resin reverse surface **82**, and a plurality of resin side surfaces **831** to **834**.

(131) As shown in FIGS. 9, 11, and 12, for example, the resin obverse surface **81** and the resin reverse surface **82** are spaced apart from each other in the z direction. The resin obverse surface **81** faces in the z2 direction, and the resin reverse surface **82** faces in the z1 direction. The control terminals **45** (first control terminals **46A** to **46E** and the second control terminals **47A** to **47D**) protrude from the resin obverse surface **81**. As shown in FIG. 10, the resin reverse surface **82** has a frame shape surrounding the bottom surface **302** of the supporting substrate **3** (lower surface of the second metal layer **33**) in plan view. The bottom surface **302** of the supporting substrate **3** is exposed from the resin reverse surface **82**, and is flush with the resin reverse surface **82**, for example. The resin side surfaces **831** to **834** are joined to the resin obverse surface **81** and the resin reverse surface **82**, and are flanked by these surfaces in the z direction. As shown in FIG. 4, the resin side surface **831** and the resin side surface **832** are spaced apart from each other in the x direction. The resin side surface **831** faces in the x1 direction, and the resin side surface **832** faces in the x2 direction. The two output terminals **44** protrude from the resin side surface **831**, and the three input terminals **41** to **43** protrude from the resin side surface **832**. As shown in FIG. 4, for example, the resin side surface **833** and the resin side surface **834** are spaced apart from each other in the y direction. The resin side surface **833** faces in the y1 direction, and the resin side surface **834** faces in the y2 direction.

(132) As shown in FIG. 4, the resin side surface **832** is formed with a plurality of recessed portions **832a**. The recessed portions **832a** are recessed in the x direction in plan view. The recessed portions **832a** include one formed between the input terminal **41** and the input terminal **42**, and one formed between the input terminal **41** and the input terminal **43** in plan view. The recessed portions **832a** are provided to increase the creepage distance between the input terminal **41** and the input terminal **42** along the resin side surface **832**, and to increase the creepage distance between the input terminal **41** and the input terminal **43** along the resin side surface **832**.

(133) As shown in FIGS. 13 and 14, the sealing resin **8** has a plurality of first protrusions **851**, a plurality of second protrusions **852**, and resin voids **86**.

(134) The first protrusions **851** protrude from the resin obverse surface **81** in the z direction. The first protrusions **851** are arranged near the four corners of the sealing resin **8** in plan view. The tip end (end in the z2 direction) of each of the first protrusions **851** is formed with a first protruding end surface **851a**. The first protruding end surfaces **851a** of the first protrusions **851** are substantially parallel to the resin obverse surface **81** and positioned on the same plane (x-y plane) as the resin obverse surface **81**. Each of the first protrusions **851** has a bottomed hollow truncated cone shape, for example. The first protrusions **851** are used for an apparatus that uses a power supply generated by the semiconductor module **A1**, and function as spacers when the

semiconductor module A1 is mounted on, for example, a control circuit board of the apparatus. Each of the first protrusions **851** has a recessed portion **851b** and an inner wall surface **851c** formed in the recessed portion **851b**. It suffices for the first protrusions **851** to have a pillar shape, preferably a columnar shape. It is preferable that each of the recessed portions **851b** have a columnar shape, and each of the inner wall surfaces **851c** have a single perfect circular shape in plan view. Each of the first protrusions **851** is an example of a “protrusion”, and each of the first protruding end surfaces **851a** is an example of a “protruding end surface”.

(135) The semiconductor module A1 may be mechanically fixed to a control circuit board or the like by, for example, a screwing method. In this case, the threads of female screws may be formed in the inner wall surfaces **851c** of the recessed portions **851b** in the first protrusions **851**. It is also possible to embed an insert nut in the recessed portion **851b** of each of the first protrusions **851**.

(136) As shown in FIG. 14, for example, the second protrusions **852** protrude from the resin obverse surface **81** in the z direction. The second protrusions **852** overlap with the control terminals **45** in plan view. The metal pins **452** of the control terminals **45** protrude from the second protrusions **852**. A part of each holder **451** (upper surface of each upper-end flange portion) is exposed from the upper end surface of each second protrusion **852**. Each of the second protrusions **852** has a truncated cone shape. The resin members **87** are provided on the second protrusions **852**.

(137) As shown in FIG. 13, each of the resin voids **86** passes from the resin obverse surface **81** to the recessed portion **201a** formed in the obverse surface **201** of the conductive substrate **2** in the z direction. Each of the resin voids **86** is formed to be tapered such that the cross-sectional area thereof decreases along the z direction from the resin obverse surface **81** to the recessed portion **201a**. A resin void edge **861** of each of the resin voids **86**, which is in contact with the obverse surface **201**, and a recess edge **201b** of each of the recessed portions **201a**, which is in contact with the obverse surface **201**, coincide with each other. The resin voids **86** are portions that are formed in a molding process described below, and in which the sealing resin **8** is not formed during the molding process.

(138) The resin members **87** are provided on the second protrusions **852** of the sealing resin **8**. The resin members **87** cover parts of the control terminals **45**, i.e., parts (upper surfaces of the upper-end flange portions) of the holders **451** that are exposed from the sealing resin **8**, and parts of the metal pins **452**. For example, the resin members **87** are made of epoxy resin, as with the sealing resin **8**, but may be made of a material different from the material of the sealing resin **8**.

(139) The resin-filling portions **88** are provided for the resin voids **86** to fill the resin voids **86**. For example, the resin-filling portions **88** are made of epoxy resin, as with the sealing resin **8**, but may be made of a material different from the material of the sealing resin **8**.

(140) The following describes a method for manufacturing the semiconductor module A1, with reference to FIGS. 21 to 29. FIG. 21 is a plan view illustrating a step of the method for manufacturing the semiconductor module A1. FIG. 22 is a schematic cross-sectional view illustrating a step of the method for manufacturing the semiconductor module A1. FIG. 23 is a plan view illustrating a step of the method for manufacturing the semiconductor module A1. FIG. 24 is a cut end view illustrating a step of the method for manufacturing the semiconductor module A1. FIG. 24 corresponds to the cross section shown in FIG. 13. Each of FIGS. 25 and 28 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module A1, and corresponds to an enlarged view of a part of the cross-section in FIG. 13. Each of FIGS. 26, 27, and 29 is a partially enlarged cross-sectional view illustrating a step of the method for manufacturing the semiconductor module A1, and corresponds to an enlarged view of a part of the cross-section in FIG. 14.

(141) First, a plurality of semiconductor elements **10**, a conductive substrate **2**, a supporting substrate **3**, a plurality of input terminals **41** to **43**, and a plurality of output terminals **44** are prepared. The configurations of the semiconductor elements **10**, the conductive substrate **2**, and the supporting substrate **3** are as described above. At the stage of preparing them, the semiconductor

elements **10**, the conductive substrate **2**, and the supporting substrate **3** are separately prepared and not bonded to each other. As shown in FIG. **21**, the conductive substrate **2**, the input terminals **41** to **43**, and the output terminals **44** are connected to each other, and may be made of the same lead frame. As shown in FIG. **21**, no recessed portion **201a** is formed in the obverse surface **201** of the conductive substrate **2**.

(142) Next, as shown in FIG. **22**, the conductive substrate **2** is placed on the supporting substrate **3** via a first conductive bonding member **71**, and the semiconductor elements **10** are placed on the conductive substrate **2** via a second conductive bonding member **72**. Then, heat is applied while the lower surface of the supporting substrate **3** and the upper surfaces of the semiconductor elements **10** are held (see the thick arrows in FIG. **22**). As a result, the semiconductor elements **10** and the conductive substrate **2** are bonded to each other by solid-phase diffusion, and the conductive substrate **2** and the supporting substrate **3** are bonded to each other by solid-phase diffusion. Specifically, the following elements are collectively bonded to each other by solid-phase diffusion: a first bonding layer **321** (supporting substrate **3**) on a first metal layer **32** and a second layer **713** (first conductive bonding member **71**); a first layer **712** (first conductive bonding member **71**) and a reverse-surface bonding layer **23** (conductive substrate **2**); a fourth layer **723** (second conductive bonding member **72**) and an obverse-surface bonding layer **22** (conductive substrate **2**); and a third layer **722** (second conductive bonding member **72**) and reverse-surface electrodes **15** of the semiconductor elements **10**. As for the conditions of the solid-phase diffusion, the heat temperature during bonding may be in the range of 200° C. to 350° C. inclusive, and the pressure applied (force for the holding) during the bonding may be in the range of 1 MPa to 100 MPa inclusive. The solid-phase diffusion bonding is assumed to be performed in the atmosphere, but it may be performed in vacuum instead. As a result, the conductive substrate **2** is bonded to the supporting substrate **3** via the first conductive bonding member **71**, and the semiconductor elements **10** are bonded to the conductive substrate **2** via the second conductive bonding member **72**. Note that the bonding between the conductive substrate **2** and the supporting substrate **3**, and the bonding between the conductive substrate **2** and the semiconductor elements **10** may be performed separately rather than collectively. However, it is more preferable to perform the bonding collectively in order to improve manufacturing efficiency.

(143) When the semiconductor elements **10** are placed on the conductive substrate **2** via the second conductive bonding member **72**, individual second conductive bonding members **72** corresponding to the respective semiconductor elements **10** may be provided as shown in FIGS. **16** and **17**. Alternatively, it is possible to provide a single second conductive bonding member **72** corresponding to the three semiconductor elements **10** shown in FIG. **16**.

(144) Next, as shown in FIG. **23**, bonding of a control terminal support **5**, bonding of a plurality of holders **451** of a plurality of control terminals **45**, bonding of a plurality of wires **731** to **735**, bonding of a plurality of first conducting members **61**, and bonding of a second conducting member **62** are performed. The bonding of these elements may be performed in any suitable order.

(145) Next, a sealing resin **8** is formed. The sealing resin **8** is formed by molding, for example. As shown in FIG. **24**, a mold **91** for a molding process is provided with pressing pins **911** as pressing members. The tip ends of the pressing pins **911** are in contact with the obverse surface **201** of the conductive substrate **2**. At this point, recessed portions **201a** are formed in the obverse surface **201** by the pressing force of the pressing pins **911** to the obverse surface **201**. The degree of recession (depth) of the recessed portions **201a** changes depending on the strength of the pressing force or the like. The pressing pins **911** in contact with the obverse surface **201** of a first conductive portion **2A** are inserted through openings **63** of the second conducting member **62**. Then, a flowable resin material is injected into a cavity space **919** of the mold **91** via a resin flow channel and a resin inlet (both not shown) in sequence. The injected flowable resin material solidifies to form the sealing resin **8**. The sealing resin **8** thus formed has first protrusions **851**, second protrusions **852**, and resin voids **86**, which are all described above, as shown in FIGS. **25** and **26**. As shown in FIG. **25**, a resin

void edge **861** of each of the resin voids **86**, which is in contact with the obverse surface **201**, and a recess edge **201b** of each of the recessed portions **201a**, which is in contact with the obverse surface **201**, coincide with each other. As shown in FIG. **26**, the upper surface of each of the holders **451** is exposed from a second protrusion **852** and flush with the upper surface of the second protrusion **852**. As is evident from FIGS. **24** and **25**, the resin voids **86** are formed by the pressing pins **911** as a result of the flowable resin material not being filled. Note that the pressing pins **911** may be movable pins. In this case, the pressing pins **911** are preferably provided in holes formed in the mold **91** and supported elastically. Each of the pressing members does not necessarily have a pin shape, and may have a block shape instead.

(146) Next, the mold **91** is opened, and a molded body is taken out, where the molded body contains the lead frame including the conductive substrate **2**, and the sealing resin **8**. Then, the sealing resin **8** is separated from the resin that has solidified at the resin flow channel and the resin inlet. In this process, one or more resin separation marks are formed at either a first position or a second position on a resin side surface **831** of the sealing resin **8** in the x1 direction. Referring to FIG. **1**, the first position may correspond to at least one of two positions each close to a respective end of the resin side surface **831** in the y direction, or at least one of the edges of the respective ends. In the case where the separation mark is formed at one of the edges of the respective ends, it may be formed at a surface formed along the edge (C chamfered portion in plan view). Such an inclined surface may be a part of the resin side surface **831** of the sealing resin **8** in the x1 direction. The second position is located between the two output terminals **44** at the resin side surface **831** shown in FIG. **1**. The resin separation mark corresponds to the position of a resin inlet of the mold **91**, and is formed by separating the sealing resin **8** from the resin that has solidified at the resin inlet. In order to prevent unevenness of the resin flow, it is preferable that the resin be injected from the central position of the mold in the y direction. In this case, a resin separation mark is formed between the two output terminals **44**.

(147) Next, as shown in FIG. **27**, metal pins **452** of the control terminals **45** are pressed into the respective holders **451**. Specifically, the metal pins **452**, each of which has a cross-sectional dimension slightly larger than the inner diameter of a tubular portion (see FIG. **26**) of each of the holders **451**, are inserted with pressure. As a result, the holders **451** and the metal pins **452** are mechanically fixed and electrically connected to each other. The holders **451** and the metal pins **452** may be electrically connected with solder, for example. Then, resin members **87** and resin-filling portions **88** are formed as shown in FIGS. **28** and **29**. The resin members **87** and the resin-filling portions **88** may be formed by potting.

(148) Next, the lead frame is cut appropriately to separate the input terminals **41** to **43** and the output terminals **44**. For each of the input terminals **41** to **43** and the output terminals **44** shown in FIG. **21**, the area near the connecting portion (portion indicated by a dashed line in FIG. **21**) between the terminal and the outer frame portion of the lead frame may be cut with a die or the like. At this point, the input terminals **41** to **43** are formed with tip surfaces **413**, **423**, and **433**, respectively, that have input-side machining marks. Each of the output terminals **44** is formed with a tip surface **443** that has an output-side machining mark. When the lead frame has tie bars that connect, in the y direction, terminals which are adjacent in the y direction, the tie bars may be cut with a die or the like. In this case, for each of the terminals, machining marks are formed on two side surfaces that face in the y direction. The semiconductor module **A1** shown in FIGS. **1** to **20** is manufactured through the steps described above.

(149) The semiconductor module **A1** is mounted on a circuit board for control, for example. The metal pins **452** are inserted into pin holes of the circuit board on which the semiconductor module **A1** is mounted, and are connected to terminals near the pin holes. The input terminals **41**, **42**, and **43** have the input-side bonding surfaces **411**, **421**, and **431**, respectively, that face in one sense (z2 direction) of the z direction. Each of the output terminals **44** has the output-side bonding surface **441** facing in one sense (z2 direction) of the z direction. The input-side bonding surfaces **411**, **421**,

431, and the output-side bonding surfaces **441** are connected with solder, for example, to the terminals of the circuit board on which the semiconductor module **A1** is mounted.

(150) The following describes the current path from the input terminal **41** to the output terminals **44** in the semiconductor module **A1** in the present embodiment. The first main circuit current flows through a path that includes the input terminal **41**, the first conductive portion **2A**, the first semiconductor elements **10A**, the first conducting members **61**, the second conductive portion **2B**, and the output terminals **44**. The first main circuit current flows along the x direction between the second obverse-surface electrodes **12** of the first semiconductor elements **10A** and the second conductive portion **2B** via the first conducting members **61**. In the second conductive portion **2B**, the first main circuit current flows along the x direction and a direction slightly inclined from the x direction between the portions to which the first conducting members **61** are bonded and the output terminals **44**.

(151) The path of a current from the output terminals **44** to the input terminal **42** and the input terminal **43** is described below. The second main circuit current flows through a path that includes the output terminals **44**, the second conductive portion **2B**, the second semiconductor elements **10B**, the second conducting member **62**, the input terminal **42**, and the input terminal **43**. The second conducting member **62**, forming the path of the second main circuit current, includes the third wiring portion **623** extending in the y direction and the first and second wiring portions **621**, **622** joined to the respective ends of the third wiring portion **623** so as to extend in the x2 direction. Thus, the second main circuit current flows through the third wiring portion **623** as well as the first wiring portion **621** and the second wiring portion **622**. Further, the path of the second main circuit current includes the two second band portions **626** disposed between the first wiring portion **621** and the second wiring portion **622** so as to extend in the x direction and also includes the first band portion **625** disposed between the first wiring portion **621** and the second wiring portion **622** so as to extend in the y direction. Thus, the second main circuit current flows through the first wiring portion **621** and the second wiring portion **622**.

(152) The second main circuit current flows between the input terminals **42**, **43**, and the second obverse-surface electrodes **12** of the second semiconductor elements **10B** via a path including the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, the two second band portions **626**, and the first band portion **625** in the second conducting member **62**. In the first wiring portion **621**, the second wiring portion **622**, and the two second band portions **626**, the second main circuit current flows along the x direction. The direction in which the first main circuit current flows is opposite from the direction in which the second main circuit current flows.

(153) The direction in which the first main circuit current flows in the first conducting members **61** is the x direction, and the direction in which the second main circuit current flows in the first wiring portion **621**, the second wiring portion **622**, and the two second band portions **626** in the second conducting member **62** is also the x direction.

(154) The following describes the operation and advantages of the semiconductor module **A1**.

(155) The semiconductor module **A1** includes the conductive substrate **2**, the input terminals **41** to **43**, the output terminals **44**, and the conducting member **6**. The conductive substrate **2** includes the first conductive portion **2A** to which the first semiconductor elements **10A** are bonded, and the second conductive portion **2B** to which the second semiconductor elements **10B** are bonded. The input terminal **41** is joined to the first conductive portion **2A**, and is electrically connected to the first semiconductor elements **10A** via the first conductive portion **2A**. The input terminal **42** and the input terminal **43** are electrically connected to the second semiconductor elements **10B** via the second conductive member **62** (conducting member **6**). The output terminals **44** are joined to the second conductive portion **2B**, and are electrically connected to the second semiconductor elements **10B** via the second conductive portion **2B**. The conducting member **6** includes the first conducting members **61** that electrically connect the first semiconductor elements **10A** and the second conductive portion **2B**, and the second conducting member **62** that electrically connects the second

semiconductor elements **10B** and the input terminals **42** and **43**. The input terminals **41** to **43** are offset in the x2 direction relative to the conductive substrate **2**, and the output terminals **44** are offset in the x1 direction relative to the conductive substrate **2**. The two input terminals **42** and **43** are located opposite from each other with the input terminal **41** therebetween in the y direction. Suppose that a semiconductor module has a configuration different from the semiconductor module **A1** in a manner such that no input terminal **43** is provided, and the input terminals **41** and **42** are arranged side by side in the y direction. In this case, variations may occur in the path of current flowing from the input terminal **41** to the output terminals **44** via the first semiconductor elements **10A**, and in the path of current flowing from the output terminals **44** to the input terminal **42** via the second semiconductor elements **10B**. In view of this, the semiconductor module **A1** includes the two input terminals **42** and **43**, and the two input terminals **42** and **43** flank the input terminal **41**. This makes it possible to reduce variations in the path of a current flowing from the input terminal **41** to the output terminals **44** via the first semiconductor elements **10A**, and to reduce variations in the path of a current flowing from the output terminals **44** to the input terminals **42** and **43** via the second semiconductor elements **10B**. As a result, the parasitic inductance components of the semiconductor module **A1** can be reduced. In other words, the semiconductor module **A1** has a package configuration preferable for reducing parasitic inductance components.

(156) In the semiconductor module **A1**, an upper arm current path and a lower arm current path overlap with each other in plan view. The upper arm current path is the path of a current flowing from the input terminal **41** to the output terminals **44** via the first conductive portion **2A**, the first semiconductor elements **10A**, the first conducting members **61**, and the second conductive portion **2B**. In the present embodiment, as seen from FIG. 5, the upper arm current path extends from the x2 direction side to the x1 direction side. The lower arm current path is the path of a current flowing from the output terminals **44** to the input terminal **42** via the second semiconductor elements **10B** and the second conductive member **62**. In the present embodiment, as seen from FIG. 5, the lower arm current path extends from the x1 direction side to the x2 direction side. With this configuration, the magnetic field generated by the current along the upper arm current path and the magnetic field generated by the current along the lower arm current path cancel each other out, thus enabling reduction of parasitic inductance components. In particular, the conducting member **6** (each of the first conducting members **61** and the second conducting member **62**) in the semiconductor module **A1** is made of a metal plate-like member, so that an area where the upper arm current path and the lower arm current path overlap with each other in plan view can be provided appropriately. In other words, the semiconductor module **A1** has a package configuration preferable for reducing parasitic inductance components.

(157) In the semiconductor module **A1**, the second conducting member **62** that forms the lower arm current path includes the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, and the fourth wiring portion **624**. The first wiring portion **621** and the second wiring portion **622** extend in the x direction, and are respectively connected to the input terminal **42** and the input terminal **43** that are arranged opposite from each other with the input terminal **41** therebetween in the y direction. The third wiring portion **623** is joined to the first wiring portion **621** and the second wiring portion **622**, extends in the y direction, and is connected to the second semiconductor elements **10B**. The fourth wiring portion **624** is joined to the first wiring portion **621** and the second wiring portion **622**, and overlaps with the first semiconductor elements **10A** in plan view. The second conductive member **62** including the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, and the fourth wiring portion **624** is spaced apart from the obverse surface **201** (conductive substrate **2**) in the z direction, and overlaps with a wide area of the obverse surface **201** in plan view. This configuration can appropriately reduce variations in the path of a current flowing from the output terminals **44** to the input terminals **42** and **43** via the second semiconductor elements **10B**, and therefore is suitable in reducing parasitic inductance components.

(158) The first semiconductor elements **10A** and the second semiconductor elements **10B** overlap with each other as viewed in the x direction. This configuration can suppress an increase in the dimension in the y direction of the conductive substrate **2** (first conductive portion **2A** and the second conductive portion **2B**) on which the first semiconductor elements **10A** and the second semiconductor elements **10B** are arranged, and can therefore reduce the size of the semiconductor module **A1**.

(159) The fourth wiring portion **624** of the second conductive member **62** has the first band portion **625** and the second band portions **626**. The first band portion **625** is joined to the first wiring portion **621** and the second wiring portion **622**, extends in the y direction, and overlaps with the first semiconductor elements **10A** in plan view. Each of the second band portions **626** is connected to the first band portion **625** and the third wiring portion **623**, and has a band shape extending in the x direction in plan view. The second band portions **626** are spaced apart from each other in the y direction and arranged substantially in parallel to each other. In plan view, one end of each band portion **626** is connected to a part of the first band portion **625**, which is located between two first semiconductor elements **10A** adjacent in the y direction, and the other end of each band portion **626** is connected to a part of the third wiring portion **623**, which is located between two second semiconductor elements **10B** adjacent in the y direction. This configuration can increase the size of the fourth wiring portion **624** (second conductive member **62**) in plan view. This is more preferable for reducing parasitic inductance components.

(160) The first band portion **625** has the protruding areas **625a** protruding in the z2 direction relative to the other areas. The protruding areas **625a** overlap with the first semiconductor elements **10A** in plan view. According to the configuration in which the first band portion **625** has the protruding areas **625a**, the first band portion **625** is prevented from making improper contact with the first conducting members **61** bonded to the first semiconductor elements **10A**.

(161) The third wiring portion **623** has the recessed areas **623a** recessed in the z1 direction relative to the other areas. The recessed areas **623a** are bonded to the respective second semiconductor elements **10B**. This configuration can increase the size of the third wiring portion **623** (second conductive member **62**) in plan view while electrically connecting the third wiring portion **623** (second conductive member **62**) and the second semiconductor elements **10B** in a suitable manner.

(162) The semiconductor module **A1** includes the conducting member **6** (first conducting members **61** and the second conducting member **62**) having the configuration described above, and further includes the first control terminals **46A** to **46E** and the second control terminals **47A** to **47D** for controlling the first semiconductor elements **10A** and the second semiconductor elements **10B**. The first control terminals **46A** to **46E** and the second control terminals **47A** to **47D** are provided on the obverse surface **201** of the conductive substrate **2** and extend along the z direction. The semiconductor module **A1** having this configuration can have a smaller size in plan view, and therefore is suitable for reducing the size in plan view while reducing parasitic inductance components.

(163) The first control terminals **46A** to **46E** are supported by the first conductive portion **2A** and offset in the x2 direction relative to the first semiconductor elements **10A**. The second control terminals **47A** to **47D** are supported by the second conductive portion **2B** and offset in the x1 direction relative to the second semiconductor elements **10B**. The first control terminals **46A** to **46E** are arranged at intervals in the y direction, and the second control terminals **47A** to **47D** are also arranged at intervals in the y direction. As such, the first control terminals **46A** to **46E** and the second control terminals **47A** to **47D** are appropriately arranged in an area corresponding to the first semiconductor elements **10A** that constitute the upper arm circuit, and in an area corresponding to the second semiconductor elements **10B** that constitute the lower arm circuit, respectively. The semiconductor module **A1** having this configuration is more preferable for downsizing while reducing parasitic inductance components.

(164) Each of the first semiconductor elements **10A** and the second semiconductor elements **10B**

has a first obverse-surface electrode **11** (gate electrode) facing in the z2 direction. The first control terminal **46A** is connected to the first obverse-surface electrodes **11** (gate electrodes) of the first semiconductor elements **10A** via the first wires **731a**. The second control terminal **47A** is connected to the first obverse-surface electrodes **11** (gate electrodes) of the second semiconductor elements **10B** via the second wires **731b**. This makes it possible to appropriately input, to the first obverse-surface electrodes **11**, a drive signal for driving the first semiconductor elements **10A** (second semiconductor elements **10B**) that have a switching function, via the first control terminal **46A** (second control terminal **47A**) and the first wires **731a** (second wires **731b**).

(165) When the semiconductor module **A1** is mounted on a circuit board, the metal pins **452** are inserted into the pin holes of the circuit board on which the semiconductor module **A1** is mounted, and are connected to terminals near the pin holes. The input terminals **41**, **42**, and **43** have the input-side bonding surfaces **411**, **421**, and **431**, respectively, that face in one sense (z2 direction) of the z direction. The output terminals **44** have the output-side bonding surfaces **441** facing in one sense (z2 direction) of the z direction. The input-side bonding surfaces **411**, **421**, **431**, and the output-side bonding surfaces **441** are connected with solder, for example, to the terminals of the circuit board on which the semiconductor module **A1** is mounted. With this configuration, the power system circuit board to which the input terminals **41** to **43** and the output terminals **44** are connected and the control system circuit board to which the metal pins **452** are connected can be arranged in separation in the z direction. This achieves the following improvements. Firstly, an improvement is made in the degree of freedom regarding the arrangement of a signal terminal in the semiconductor module **A1**. Secondly, an improvement is made in the degree of freedom regarding the routing and length of a signal wire in the semiconductor module **A1**. Thirdly, an improvement is made in the degree of freedom regarding the arrangement of a circuit board by a user when the semiconductor module **A1** is used.

(166) In the semiconductor module **A1**, the control terminals **45** protrude from the resin obverse surface **81** and extend along the z direction. In a configuration different from that of the semiconductor module **A1**, the control terminals **45** may be arranged to extend along a plane (x-y plane) perpendicular to the z direction. This configuration has a limit to the size reduction in plan view. Accordingly, as in the semiconductor module **A1**, the control terminals **45** can be arranged to extend along the z direction, so that the size of the semiconductor module **A1** can be reduced in plan view. In other words, the semiconductor module **A1** has a package configuration preferable for the size reduction in plan view.

(167) In the semiconductor module **A1** of the present embodiment, the control terminal support **5** is provided between the control terminals **45** and the obverse surface **201** (conductive substrate **2**). The control terminal support **5** has the insulating layer **51**, and the control terminals **45** are supported by the conductive substrate **2** via the control terminal support **5**. The configuration with the control terminal support **5** can support the control terminals **45** on the conductive substrate **2** appropriately while maintaining insulation from the conductive substrate **2**.

(168) The control terminal support **5** has a layup structure in which the insulating layer **51**, the first metal layer **52**, and the second metal layer **53** are stacked on each other. The control terminals **45** are bonded to the first metal layer **52**, which is formed as the upper surface of the control terminal support **5**, via the conductive bonding member **459**. According to the configuration, the control terminals **45** can be electrically bonded to the control terminal support **5** (first metal layer **52**) while utilizing an existing layup structure (e.g., DBC substrate) as the control terminal support **5**.

(169) Each of the semiconductor elements **10** has an element obverse surface **101** facing in the z2 direction, and an element reverse surface **102** facing in the z1 direction. A first obverse-surface electrode **11** (gate electrode) is provided on the element obverse surface **101**. The first obverse-surface electrode **11** of each of the semiconductor elements **10** and the first metal layer **52** (first portion **521**) are connected by a wire **731** that is electrically conductive. This makes it possible to input a drive signal for driving the semiconductor elements **10** having a switching function to the

first obverse-surface electrodes **11** appropriately, via the control terminals **45**, the first metal layer **52**, and the wires **731**.

(170) Each of the control terminals **45** includes a holder **451** and a metal pin **452**. The holder **451** is made of a conductive material, and includes a tubular portion. The metal pin **452** is a rod-like member extending in the z direction, and is pressed into the holder **451**. A part (the upper surface of the upper-end flange portion) of the holder **451** is exposed from the sealing resin **8**. According to this configuration, the sealing resin **8** is formed (by molding) such that the holder **451** is covered with the sealing resin **8** except a part (upper end surface) of the holder **451**, and the upper end surface of the holder **451** is exposed from the sealing resin **8**. This makes it possible to insert the metal pin **452** into the holder **451** after the sealing resin **8** is formed. Accordingly, with the configuration in which the control terminals **45** include the holders **451** and the metal pins **452**, it is possible to avoid complexity of the mold **91** for a molding process. For this reason, this configuration is suitable for efficiently manufacturing the semiconductor module **A1**.

(171) The semiconductor module **A1** of the present embodiment includes the resin members **87** bonded to the sealing resin **8**. The resin members **87** cover parts (upper surfaces of the upper-end flange portions) of the holders **451** that are exposed from the sealing resin **8**, and parts of the metal pins **452**. This configuration prevents foreign matter from entering the connecting portions between the holders **451** and the metal pins **452**. The semiconductor module **A1** having the above configuration is preferable in terms of durability and reliability.

(172) The sealing resin **8** has the second protrusions **852** protruding from the resin obverse surface **81**. The second protrusions **852** surround the respective control terminals **45** in plan view. The metal pins **452** of the control terminals **45** protrude from the second protrusions **852**. The resin members **87** are provided on the second protrusions **852**. According to this configuration, the creepage distance between adjacent control terminals **45** along the resin obverse surface **81** can be increased. This is preferable for increasing the withstand voltage of the adjacent control terminals **45**.

(173) The conductive substrate **2** includes the first conductive portion **2A** and the second conductive portion **2B** that are spaced apart from each other in the x direction. The first conductive portion **2A** is offset in the x2 direction relative to the second conductive portion **2B**. The semiconductor elements **10** include the first semiconductor elements **10A** bonded to the first conductive portion **2A**, and the second semiconductor elements **10B** bonded to the second conductive portion **2B**. The control terminals **45** include the first control terminals **46A** to **46E**, and the second control terminals **47A** to **47D**. The first control terminals **46A** to **46E** are supported by the first conductive portion **2A**, and arranged between the first semiconductor elements **10A** and the input terminals **41**, **42**, etc., in the x direction. The second control terminals **47A** to **47D** are provided between the second semiconductor elements **10B** and the output terminals **44** in the x direction. With this configuration, the control terminals **45** (the first control terminals **46A** to **46E**, and the second control terminals **47A** to **47D**) are appropriately arranged in an area corresponding to the first semiconductor elements **10A** that constitute the upper arm circuit, and in an area corresponding to the second semiconductor elements **10B** that constitute the lower arm circuit. The configuration is preferable for downsizing the semiconductor module **A1**.

(174) The sealing resin **8** has the first protrusions **851** protruding from the resin obverse surface **81**. The tip end of each of the first protrusions **851** is formed with a first protruding end surface **851a**. The first protruding end surfaces **851a** of the first protrusions **851** are substantially parallel to the resin obverse surface **81** and positioned on the same plane (x-y plane) as the resin obverse surface **81**. With this configuration, it is possible, in an apparatus that uses a power supply generated by the semiconductor module **A1**, to provide a predetermined gap between the surface of a control circuit board on which the semiconductor module **A1** is mounted and the resin obverse surface **81**. In this way, even when various functional components are mounted on a surface of the control circuit board that faces the semiconductor module **A1**, the functional components do not make improper

contact with the sealing resin **8**.

(175) The semiconductor module **A1** includes the conductive substrate **2** to which the semiconductor elements **10** are bonded. With this configuration, the heat generated by energization of the semiconductor elements **10** is transferred from the semiconductor elements **10** to the conductive substrate **2** and diffused at the conductive substrate **2**. As such, the semiconductor module **A1** has a package configuration preferable for improving the heat dissipation property of the semiconductor elements **10**.

(176) In the semiconductor module **A1**, the conductive substrate **2** and the supporting substrate **3** are bonded to each other via the first conductive bonding member **71**. The first conductive bonding member **71** includes the first layer **712** and the second layer **713**. The first layer **712** is bonded to the conductive substrate **2** by the solid-phase diffusion of metal, and is in direct contact with the conductive substrate **2** at the bonding interface. The second layer **713** is bonded to the supporting substrate **3** by the solid-phase diffusion of metal, and is in direct contact with the supporting substrate **3** at the bonding interface. This configuration can increase the bonding strength between the conductive substrate **2** and the supporting substrate **3** as compared to the case where the conductive substrate **2** and the supporting substrate **3** are bonded by a bonding material such as solder. Accordingly, the semiconductor module **A1** has a package configuration preferable for suppressing the peeling between the conductive substrate **2** and the support substrate **3**.

(177) In the semiconductor module **A1**, the semiconductor elements **10** and the conductive substrate **2** are bonded to each other via the second conductive bonding member **72**. The second conductive bonding member **72** includes the third layer **722** and the fourth layer **723**. The third layer **722** is bonded to the semiconductor elements **10** (reverse surface electrodes **15**) by the solid-phase diffusion of metal, and is in direct contact with the semiconductor elements **10** at the bonding interface. The fourth layer **723** is bonded to the conductive substrate **2** by the solid-phase diffusion of metal, and is in direct contact with the conductive substrate **2** at the bonding interface. This configuration can increase the bonding strength between the semiconductor elements **10** and the conductive substrate **2** as compared to the case where the semiconductor elements **10** and the conductive substrate **2** are bonded by a bonding material such as solder. Accordingly, the semiconductor module **A1** has a package configuration preferable for suppressing the peeling between the semiconductor elements **10** and the conductive substrate **2**.

(178) In the semiconductor module **A1** of the present embodiment, the Young's modulus of the first base layer **711** in the first conductive bonding member **71** is smaller than the Young's modulus of the material of each of the first layer **712** and the second layer **713**. According to the configuration, when the first conductive bonding member **71** is bonded to the conductive substrate **2** and the supporting substrate **3** by solid-phase diffusion, the stress is alleviated by the relatively soft first base layer **711**, and the bonding boundary portion is thereby smoothed. As a result, the first layer **712** and the conductive substrate **2**, as well as the second layer **713** and the supporting substrate **3**, are more firmly bonded by solid-phase diffusion.

(179) In the present embodiment, the first base layer **711** is thicker than each of the first layer **712** and the second layer **713**. Accordingly, when bonding by solid-phase diffusion is performed, the pressing force acting on the boundary portion between the first layer **712** and the conductive substrate **2** (reverse-surface bonding layer **23**) and on the boundary portion between the second layer **713** and the supporting substrate **3** (first bonding layer **321**) is made more uniform. As a result, the first layer **712** and the conductive substrate **2**, as well as the second layer **713** and the supporting substrate **3**, can be in a stronger conductive bonding state.

(180) The material of each of the first layer **712** and the second layer **713** contains silver. With this composition, when bonding by solid-phase diffusion is performed with the first conductive bonding member **71**, oxidation of the first layer **712** and the second layer **713** is suppressed, thus enabling excellent solid-phase diffusion bonding. The reverse-surface bonding layer **23** and the first bonding layer **321**, which are bonded to the first layer **712** and the second layer **713** respectively, also

contain silver, thus enabling better solid-phase diffusion bonding.

(181) In the present embodiment, the Young's modulus of the second base layer **721** in the second conductive bonding member **72** is smaller than the Young's modulus of the material of each of the third layer **722** and the fourth layer **723**. According to the configuration, when the second conductive bonding member **72** is bonded to the semiconductor elements **10** (reverse-surface electrodes **15**) and the conductive substrate **2** by solid-phase diffusion, the stress is alleviated by the relatively soft second base layer **721**, and the bonding boundary portion is thereby smoothed. As a result, the third layer **722** and the semiconductor elements **10** (reverse-surface electrodes **15**), as well as the fourth layer **723** and the conductive substrate **2**, are more firmly bonded by solid-phase diffusion.

(182) In the present embodiment, the second base layer **721** is thicker than each of the third layer **722** and the fourth layer **723**. Accordingly, when bonding by solid-phase diffusion is performed, the pressing force acting on the boundary portion between the third layer **722** and the semiconductor elements **10** (reverse-surface electrodes **15**) and on the boundary portion between the fourth layer **723** and the conductive substrate **2** (obverse-surface bonding layers **22**) is made more uniform. As a result, the third layer **722** and the semiconductor elements **10** (reverse-surface electrodes **15**), as well as the fourth layer **723** and the conductive substrate **2**, can be in a stronger conductive bonding state.

(183) The material of each of the third layer **722** and the fourth layer **723** contains silver. With this material composition, when bonding by solid-phase diffusion is performed with the second conductive bonding member **72**, oxidation of the third layer **722** and the fourth layer **723** is suppressed, thus enabling excellent solid-phase diffusion bonding. The reverse-surface electrodes **15** and the obverse-surface bonding layers **22**, which are bonded to the third layer **722** and the fourth layer **723** respectively, contain silver, thus enabling better solid-phase diffusion bonding.

(184) The first conductive bonding member **71** has a configuration where the first layer **712** and the second layer **713**, which are Ag plating layers, are formed on the surfaces (both surfaces) of the first base layer **711**, which is made of a sheet material containing **A1**. Similarly, the second conductive bonding member **72** has a configuration where the third layer **722** and the fourth layer **723**, which are Ag plating layers, are formed on the surfaces (both surfaces) of the second base layer **721**, which is made of a sheet material containing **A1**. With this configuration, the first conductive bonding member **71** and the second conductive bonding member **72** can be easily prepared.

(185) In the semiconductor module **A1**, the second conducting member **62** is formed with the openings **63**. The openings **63** overlap with the obverse surface **201** (conductive substrate **2**) in plan view, and do not overlap with the semiconductor elements **10** in plan view. With this configuration, during a molding step (step for forming the sealing resin **8**) in the process for manufacturing the semiconductor module **A1**, the pressing pins **911** of the mold **91** can be inserted into the openings **63**. This allows the pressing pins **911** to press the conductive substrate **2** without interfering with the second conducting member **62**, thus suppressing the warpage of the supporting substrate **3** to which the conductive substrate **2** is bonded. The warpage occurs, for example, such that the outer sides of the supporting substrate **3** in the y direction are positioned more upward than the center thereof in the y direction. If warpage occurs on the supporting substrate **3**, the bonding strength between the conductive substrate **2** and the supporting substrate **3** may be lowered. Furthermore, during a molding process, a part of the sealing resin **8** may be formed on the bottom surface **302** due to resin leakage, causing a bonding failure of a heat dissipating member (e.g., heat sink) that can be bonded to the bottom surface **302**. Accordingly, the semiconductor module **A1** has a package configuration that is preferable for improving the bonding strength between the conductive substrate **2** and the supporting substrate **3** by suppressing the warpage of the supporting substrate **3**, and that is also preferable for suppressing the leakage of the sealing resin **8** to an unintended location.

(186) The conductive substrate **2** includes the first conductive portion **2A** to which the first semiconductor elements **10A** are bonded, and the second conductive portion **2B** to which the second semiconductor elements **10B** are bonded. The first conductive portion **2A** and the second conductive portion **2B** are spaced apart from each other in the x direction, and the first conductive portion **2A** is offset in the x2 direction relative to the second conductive portion **2B**. The second conducting member **62** is connected to the second semiconductor elements **10B** and the input terminals **42** and **43**, and the openings **63** in the second conducting member **62** overlap with the obverse surface **201** of the first conductive portion **2A** in plan view. With this configuration, even when the second conducting member **62** is designed to have a large size in plan view, the pressing pins **911** of the mold **91** can press the conductive substrate **2** without interfering with the second conducting member **62** during the formation (during the molding process) of the sealing resin **8**. Note that the parasitic resistance components of the second conducting member **62** (conductive member **6**) that forms the path of the main circuit current can be suppressed by increasing the size of the second conducting member **62** in plan view.

(187) The second conducting member **62** includes the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, and the fourth wiring portion **624**. The first wiring portion **621** and the second wiring portion **622** extend in the x direction, and are respectively connected to the input terminal **42** and the input terminal **43** that are arranged opposite from each other with the input terminal **41** therebetween in the y direction. The third wiring portion **623** is joined to the first wiring portion **621** and the second wiring portion **622**, extends in the y direction, and is connected to the second semiconductor elements **10B**. The openings **63** are formed in the areas of the first wiring portion **621** and the second wiring portion **622** that are offset in the x2 direction. As such, the openings **63** are provided near two corners of the conductive substrate **2** (first conductive portion **2A**) at respective outer sides of the conductive substrate **2** in the y direction. Accordingly, the openings **63** are provided near two corners of the supporting substrate **3** supporting the conductive substrate **2** (first conductive portion **2A**) at the respective outer sides of the supporting substrate **3** in the y direction. The configuration as described above allows the size of the second conducting member **62** to be relatively large in plan view and, during the formation of the sealing resin **8** (molding process), areas near the two corners of the conductive substrate **2** (first conductive portion **2A**) at the respective outer sides of the conductive substrate **2** in the y direction can be pressed with the pressing pins **911** of the mold **91** which are inserted into the openings **63**. As described above, the warpage of the supporting substrate **3** to which the conductive substrate **2** is bonded occurs such that the outer sides of the supporting substrate **3** in the y direction are positioned more upward than the center thereof in the y direction. However, the configuration described above can effectively suppress the warpage of the supporting substrate **3** during the molding process.

(188) In the present embodiment, the conducting member **6** (the first conducting members **61** and the second conducting member **62**) is made of a metal plate-like member. This facilitates formation of the openings **63** in the second conducting member **62**. Furthermore, the conducting member **6** (the first conducting members **61** and the second conducting member **62**) made of a metal plate-like member can easily adapt to various shapes and sizes, and can increase the reliability of a bonding portion with another component by securing a sufficient bonding area with the other component.

(189) Parts of the obverse surface **201** of the conductive substrate **2** (first conductive portion **2A**) overlap with the openings **63** in plan view and are formed with the recessed portions **201a**. The recessed portions **201a** are marks left by the pressing pins **911** applying a pressing force to the obverse surface **201** during the molding process. In the present embodiment, it is possible to devise an arrangement of the second conducting member **62** and the openings **63** formed therein, so that during the molding process, appropriate parts of the conductive substrate **2** (first conductive portion **2A**) can be pressed with the pressing pins **911** while avoiding interference with functional elements

such as the semiconductor elements **10**.

(190) The sealing resin **8** is formed with the resin voids **86** passing from the resin obverse surface **81** to the recessed portions **201a**. Each of the resin voids **86** is tapered such that the cross-sectional area thereof decreases from the resin obverse surface **81** to the recessed portion **201a**. The resin voids **86** are formed during a molding process (when the sealing resin **8** is formed). After the molding, the surfaces of the recessed portions **201a** in the obverse surface **201** of the conductive substrate **2** are exposed from the sealing resin **8**. In the present embodiment, the resin-filling portions **88** are provided for the resin voids **86** to fill the resin voids **86**. This configuration can prevent foreign matter (such as moisture) from entering the recessed portions **201a** exposed from the sealing resin **8**. The semiconductor module **A1** having the above configuration is preferable in terms of durability and reliability.

(191) In the present embodiment, the openings **63** in the second conducting member **62** (conducting member **6**) are through-holes that penetrate through in the z direction. This configuration can prevent a deviation of the current path caused by forming the openings **63** in the second conducting member **62** (conducting member **6**) that forms the path of a main circuit current.

(192) The semiconductor module **A1** includes the conducting member **6**. The conducting member **6** forms the path of a main circuit current switched by the semiconductor elements **10**. The conducting member **6** includes the first conducting members **61** connected to the first semiconductor elements **10A**, and the second conducting member **62** connected to the second semiconductor elements **10B**. The conducting member **6** (each of the first conducting members **61** and the second conducting member **62**) is made of a metal plate-like member. The main circuit current described above may have a relatively large value. In this case, it is preferable to suppress the parasitic resistance components in the conducting member **6** that forms the path of the main circuit current in order to reduce the power consumption of the semiconductor module **A1**.

Accordingly, in the semiconductor module **A1**, the conducting member **6** is made of a metal plate-like member instead of a bonding wire as described above to suppress the parasitic resistance components of the conducting member **6**. In other words, the semiconductor module **A1** has a package configuration preferable for suppressing the parasitic resistance components.

(193) In the semiconductor module **A1**, each of the first semiconductor elements **10A** has a rectangular shape in plan view, and the four corners of each of the first semiconductor elements **10A** in plan view do not overlap with the second conducting member **62**. According to this configuration, during the manufacturing process of the semiconductor module **A1**, it is possible to conduct visual inspection before forming the sealing resin **8** so as to check whether the first semiconductor elements **10A** are properly bonded. In other words, the semiconductor module **A1** allows for visual inspection regarding the bonding state of the first semiconductor elements **10A** during the manufacturing process (e.g., the stage shown in FIG. 23). This makes it possible to determine whether the first semiconductor elements **10A** are properly bonded. For example, it is possible to measure respective distances to the four corners of each first semiconductor element **10A** by a laser ranging method, and determine that the first semiconductor element **10A** is properly bonded if the difference between the distances to the four corners is small. As described above, the semiconductor module **A1** can conduct visual inspection during the manufacturing process, and therefore has package configuration preferable for improving reliability. During the visual inspection, it is sufficient if at least three of the four corners of each of the first semiconductor elements **10A** are visible in plan view. For this reason, it is sufficient if three corners of each of the first semiconductor elements **10A** do not overlap with the second conducting member **62**. Similarly, as shown in FIG. 5, four corners of each of the second semiconductor elements **10B** do not overlap with the second conducting member **62**. Accordingly, during the manufacturing process of the semiconductor module **A1**, it is possible to conduct visual inspection before forming the sealing resin **8** so as to check whether the second semiconductor elements **10B** are properly bonded. The visual inspection may be automatic visual inspection that uses image-capturing and image

processing.

(194) The second conducting member **62** includes the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, and the fourth wiring portion **624**. The first wiring portion **621** and the second wiring portion **622** extend in the x direction, and are respectively connected to the input terminal **42** and the input terminal **43** that are arranged opposite from each other with the input terminal **41** therebetween in the y direction. The third wiring portion **623** is joined to the first wiring portion **621** and the second wiring portion **622**, extends in the y direction, and is connected to the second semiconductor elements **10B**. The fourth wiring portion **624** is joined to the first wiring portion **621** and the second wiring portion **622**. The fourth wiring portion **624** is offset in the x2 direction relative to the third wiring portion **623**, and overlaps with the first semiconductor elements **10A** in plan view. The second conductive member **62** including the first wiring portion **621**, the second wiring portion **622**, the third wiring portion **623**, and the fourth wiring portion **624** overlaps with a wide area of the obverse surface **201** in plan view, and has a relatively large size in plan view. Increasing the size of the second conducting member **62** in plan view is preferable in terms of suppressing the parasitic resistance components of the second conducting member **62** (conductive member **6**) that forms the path of the main circuit current.

(195) Each of the first semiconductor elements **10A** has a first side **191**, a second side **192**, a third side **193**, and a fourth side **194** in plan view. The first side **191** and the second side **192** extend in the y direction. The first side **191** is an edge located in the x2 direction in plan view, and the second side **192** is an edge located in the x1 direction in plan view. The third side **193** and the fourth side **194** extend in the x direction. The third side **193** is an edge located in the y2 direction in plan view, and the fourth side **194** is an edge located in the y1 direction in plan view. Since each of the first semiconductor elements **10A** has a rectangular shape in plan view, the four corners formed by the first side **191**, the second side **192**, the third side **193**, and the fourth side **194** are generally right-angled in plan view. The fourth wiring portion **624** (the first band portion **625**) of the second conducting member **62** has a first edge **627** and a second edge **628**. The first edge **627** is an edge of the fourth wiring portion **624** located in the x2 direction, and is offset in the x1 direction relative to the first side **191** in plan view. The first edge **627** extends at least from the third side **193** to the fourth side **194** in the y direction. As such, two corners **171** and **172** of each first semiconductor element **10A** in the x2 direction do not overlap with the second conducting member **62** in plan view. The second edge **628** is an edge of the fourth wiring portion **624** (first band portion **625**) located in the x1 direction, and is offset in the x2 direction relative to the second side **192** in plan view. The second edge **628** extends at least from the third side **193** to the fourth side **194** in the y direction. As such, two corners **173** and **174** of each first semiconductor element **10A** in the x1 direction do not overlap with the second conducting member **62** in plan view. With this configuration, the four corners of each of the first semiconductor elements **10A** in plan view do not overlap with the second conducting member **62** while the size of the second conducting member **62** in plan view is increased by providing the fourth wiring portion **624** with areas that overlap with the first semiconductor elements **10A** in plan view. This makes it possible to effectively suppress the parasitic resistance components of the second conducting member **62** (conducting member **6**), and to conduct visual inspection to check the bonding state of the first semiconductor elements **10A** during the manufacturing process of the semiconductor module **A1**.

(196) The fourth wiring portion **624** (first band portion **625**) has the protruding areas **625a** protruding in the z2 direction relative to the other areas. The protruding areas **625a** overlap with the first semiconductor elements **10A** in plan view. According to the configuration in which the fourth wiring portion **624** has the protruding areas **625a**, the fourth wiring portion **624** is prevented from making improper contact with the first conducting members **61** bonded to the first semiconductor elements **10A**.

(197) The third wiring portion **623** has the recessed areas **623a** recessed in the z1 direction relative to the other areas. The recessed areas **623a** are bonded to the respective second semiconductor

elements **10B**. This configuration can increase the size of the third wiring portion **623** (second conductive member **62**) in plan view while electrically connecting the third wiring portion **623** (second conductive member **62**) and the second semiconductor elements **10B** in a suitable manner. (198) The first semiconductor elements **10A** and the second semiconductor elements **10B** overlap with each other as viewed in the x direction. This configuration can suppress an increase in the dimension in the y direction of the conductive substrate **2** (first conductive portion **2A** and the second conductive portion **2B**) on which the first semiconductor elements **10A** and the second semiconductor elements **10B** are arranged, and can therefore reduce the size of the semiconductor module **A1**.

(199) The semiconductor module **A1** includes the conductive substrate **2**, the two input terminals **41** and **42** (or the two input terminals **41** and **43**), the output terminals **44**, and the conducting member **6**. The conductive substrate **2** includes the first conductive portion **2A** and the second conductive portion **2B** aligned in the x direction in plan view. The first semiconductor elements **10A** are electrically bonded to the first conductive portion **2A**. The second semiconductor elements **10B** are electrically bonded to the second conductive portion **2B**. The first semiconductor elements **10A** and the second semiconductor elements **10B** are arranged at intervals in the y direction. The two input terminals **41** and **42** (or the two input terminals **41** and **43**) are offset in the x2 direction relative to the first conductive portion **2A**. The input terminal **41** is a positive electrode, and is connected to the first conductive portion **2A**. The input terminal **42** (or the input terminal **43**) is a negative electrode. The output terminals **44** are offset in the x1 direction relative to the second conductive portion **2B**. The conducting member **6** includes the first conducting members **61** connected to the first semiconductor elements **10A** and the second conductive portion **2B**, and the second conducting member **62** connected to the second semiconductor elements **10B** and the input terminal **42** (or the input terminal **43**). According to this configuration, the path of the main circuit current switched by the semiconductor elements **10** (the first semiconductor elements **10A** and the second semiconductor elements **10B**) is formed along the x direction in plan view, and the axis of symmetry (see an auxiliary line **L1** in FIG. 5) in the planar structure of the semiconductor module **A1** extends along the y direction in plan view. In other words, the axis of symmetry and the path of the main circuit current are perpendicular to each other. This reduces the difference in the current path to the first semiconductor elements **10A** and the second semiconductor elements **10B** regarding the main circuit current inputted from the two input terminals **41** and **42** (or the two input terminals **41** and **43**) and outputted from the output terminals **44**. Accordingly, it is possible to suppress variations in parasitic inductance components and variations in current in the semiconductor module **A1**. Accordingly, the semiconductor module **A1** has a package configuration preferable for equalizing the parasitic inductance components in the path of the main circuit current and for equalizing the amount of current to the semiconductor elements **10**.

(200) The first semiconductor elements **10A** and the second semiconductor elements **10B** are spaced apart in the x direction. The first semiconductor elements **10A** and the second semiconductor elements **10B** are aligned along the y direction. Accordingly, the direction in which the semiconductor elements **10** are aligned is perpendicular to the direction in which the first main circuit current or the second main circuit current flows. In this way, when a plurality of switching elements are connected in parallel for use as in the present embodiment, the difference in the length of the path of the first main circuit current between the three first semiconductor elements **10A** can be reduced. This makes it possible to suppress the parasitic resistance components in the conducting member **6** that forms the path of the main circuit current.

(201) The area in which the first main circuit current flows and the area in which the second main circuit current flows overlap with each other in plan view. In other words, the second conducting member **62**, which connects the output terminals **44** to the input terminals **42** and **43** that are negative electrode terminals to let the second main circuit current flow, is arranged above the area (the first conductive portion **2A**, the first conducting members **61**, and the second conductive

portion 2B) in which the first main circuit current flows. The direction in which the first main circuit current flows is opposite from the direction in which the second main circuit current flows. With the arrangement described above, the magnetic field generated by the first main circuit current and the magnetic field generated by the second main circuit current cancel each other out, thus enabling reduction of inductance.

(202) The semiconductor module A1 of the present embodiment includes the two input terminals 42 and 43. The input terminals 42 and 43 are negative electrodes and flank the input terminal 41 in the y direction. The two input terminals 42 and 43 are connected to the second conducting member 62. This configuration can further reduce variations in the path of a current flowing from the output terminals 44 to the input terminals 42 and 43 via the second semiconductor elements 10B and the second conducting member 62.

(203) In the semiconductor module A1, the second conducting member 62 includes the first wiring portion 621, the second wiring portion 622, the third wiring portion 623, and the fourth wiring portion 624. The first wiring portion 621 and the second wiring portion 622 extend in the x direction, and are respectively connected to the input terminal 42 and the input terminal 43 that are arranged opposite from each other with the input terminal 41 therebetween in the y direction. The third wiring portion 623 is joined to the first wiring portion 621 and the second wiring portion 622, extends in the y direction, and is connected to the second semiconductor elements 10B. The fourth wiring portion 624 is offset in the x2 direction relative to the third wiring portion 623, and is joined to the first wiring portion 621, the second wiring portion 622, and the third wiring portion 623. The second conductive member 62 including the first wiring portion 621, the second wiring portion 622, the third wiring portion 623, and the fourth wiring portion 624 overlaps with a wide area of the obverse surface 201 in plan view, and has a relatively large size in plan view. This configuration can appropriately reduce variations in the path of a current flowing from the output terminals 44 to the input terminals 42 and 43 via the second semiconductor elements 10B and the second conducting member 62. Accordingly, the semiconductor module A1 of the present embodiment is more preferable for equalizing the parasitic inductance components in the path (second conducting member 62) of the main circuit current and for equalizing the amount of current to the semiconductor elements 10B.

(204) The fourth wiring portion 624 is joined to the first wiring portion 621 and the second wiring portion 622, and overlaps with the first semiconductor elements 10A in plan view. The fourth wiring portion 624 (first band portion 625) has the protruding areas 625a protruding in the z2 direction relative to the other areas. The protruding areas 625a overlap with the first semiconductor elements 10A in plan view. This configuration can increase the size of the fourth wiring portion 624 (second conductive member 62) in plan view, and can prevent the fourth wiring portion 624 from making improper contact with the first conducting members 61 bonded to the first semiconductor elements 10A.

(205) The first semiconductor elements 10A and the second semiconductor elements 10B overlap with each other as viewed in the x direction. This configuration can suppress an increase in the dimension in the y direction of the conductive substrate 2 (first conductive portion 2A and the second conductive portion 2B) on which the first semiconductor elements 10A and the second semiconductor elements 10B are arranged, and can therefore reduce the size of the semiconductor module A1.

(206) FIGS. 30 to 32 illustrate a semiconductor module according to a second embodiment. A semiconductor module A2 of the present embodiment is different from the semiconductor module A1 of the first embodiment in the configuration of the second conducting member 62.

(207) The present embodiment is different from the first embodiment in the area occupied by the fourth wiring portion 624 of the second conducting member 62. Specifically, the dimension of the first band portion 625 in the x direction is larger than that of the semiconductor module A1. As shown in FIGS. 31 and 32, the second edge 628 of the first band portion 625 is offset in the x1

direction as compared to the second edge **628** in the semiconductor module **A1**. As shown in FIG. **32**, the second edge **628** is offset in the x1 direction relative to the second sides **192** of the first semiconductor elements **10A** in plan view. As such, two corners of each first semiconductor element **10A** in the x1 direction overlap with the second conducting member **62** (first band portion **625**) in plan view.

(208) The semiconductor module **A2** of the present embodiment has the same advantages as the semiconductor module **A1** of the first embodiment. Furthermore, in the semiconductor module **A2**, the first band portion **625** (second conducting member **62**) of the fourth wiring portion **624** can have a larger size in plan view. This is more preferable for reducing parasitic inductance components.

(209) FIGS. **33** and **34** illustrate a semiconductor module according to a third embodiment. A semiconductor module **A3** of the present embodiment is different from the semiconductor module **A1** of the first embodiment mainly in the configuration of the second conducting member **62**.

(210) Unlike the above embodiments, the second conducting member **62** of the semiconductor module **A3** does not have any openings **63**. The mold **91** used to form the sealing resin **8** (by molding) in the manufacturing of the semiconductor module **A3** is not provided with the pressing pins **911**. Accordingly, as shown in FIG. **34**, the sealing resin **8** is not formed with the resin voids **86**, and the obverse surface **201** of the conductive substrate **2** (the first conductive portion **2A** and the second conductive portion **2B**) is not formed with the recessed portions **201a**. Since the sealing resin **8** is not formed with the resin voids **86**, the semiconductor module **A3** of the present embodiment does not have any resin-filling portions **88** which are used to fill the resin voids **86** in the first embodiment.

(211) The semiconductor module **A3** of the present embodiment has the same advantages as the semiconductor module **A1** of the first embodiment.

(212) The semiconductor module according to the present disclosure is not limited to the above embodiments. Various design changes can be made to the specific configurations of the elements of the semiconductor module in the present disclosure.

(213) The materials, etc., of the layers that constitute the first conductive bonding member or the second conductive bonding member of the present disclosure are not limited to those in the above embodiments. In the above embodiments, the conductive substrate **2** and the semiconductor elements **10** (the first semiconductor elements **10A** and the second semiconductor elements **10B**) are bonded with the second conductive bonding member **72** therebetween by solid-phase diffusion bonding. Alternatively, the conductive substrate **2** and the semiconductor elements **10** may be bonded with another conductive bonding member such as solder.

(214) The present disclosure includes the configurations defined in the following clauses.

(215) Clause 1.

(216) A semiconductor module comprising: a supporting substrate having a supporting surface that faces in one sense of a thickness direction; a conductive substrate having an obverse surface and a reverse surface that face away from each other in the thickness direction, the conductive substrate being supported by the supporting substrate such that the reverse surface faces the supporting surface; a first conductive bonding member provided between the supporting substrate and the conductive substrate and electrically bonding the supporting substrate and the conductive substrate; and a semiconductor element electrically bonded to the obverse surface and having a switching function, wherein the first conductive bonding member includes a first base layer made of metal, a first layer, and a second layer, the first layer being provided between the first base layer and the conductive substrate and bonded in direct contact with the conductive substrate at a bonding interface with the conductive substrate, the second layer being provided between the first base layer and the supporting substrate and bonded in direct contact with the supporting substrate at a bonding interface with the supporting substrate.

Clause 2.

(217) The semiconductor module according to clause 1, further comprising: a sealing resin having a resin obverse surface facing the same side as the obverse surface, a resin reverse surface facing an opposite side from the resin obverse surface, and a resin side surface connected to the resin obverse surface and the resin reverse surface and flanked by the resin obverse surface and the resin reverse surface in the thickness direction, the sealing resin covering at least a part of the supporting substrate, the conductive substrate, the first conductive bonding member, and the semiconductor element; an input terminal electrically connected to the semiconductor element and protruding from the resin side surface; an output terminal electrically connected to the semiconductor element, and protruding from the resin side surface; and a control terminal for controlling the semiconductor element, wherein the input terminal has an input-side bonding surface facing in the one sense of the thickness direction, the output terminal has an output-side bonding surface facing in the one sense of the thickness direction, and the control terminal is arranged on the obverse surface and extends along the thickness direction.

Clause 3.

(218) The semiconductor module according to clause 2, wherein a Young's modulus of the first base layer is smaller than a Young's modulus of each of the first layer and the second layer.

(219) Clause 4.

(220) The semiconductor module according to clause 3, wherein the first base layer is thicker than each of the first layer and the second layer.

(221) Clause 5.

(222) The semiconductor module according to clause 4, wherein each of the first layer and the second layer contains silver.

(223) Clause 6.

(224) The semiconductor module according to clause 5, wherein the first base layer is made of a sheet material containing aluminum, and each of the first layer and the second layer is a silver plating layer stacked on a surface of the first base layer.

Clause 7.

(225) The semiconductor module according to any of clauses 2 to 6, wherein the supporting substrate has an insulating layer, a first metal layer stacked on a surface of the insulating layer in the one sense of the thickness direction, and a first bonding layer made of metal that includes the supporting surface and is stacked on the first metal layer, and the first bonding layer and the second layer are bonded in direct contact with each other at a bonding interface.

Clause 8.

(226) The semiconductor module according to any of clauses 2 to 7, wherein the conductive substrate has a base member made of a metal plate, and a reverse-surface bonding layer made of metal that includes the reverse surface and is stacked on a surface of the base member in another sense of the thickness direction, and the reverse-surface bonding layer and the first layer are bonded in direct contact with each other at a bonding interface.

Clause 9.

(227) The semiconductor module according to clause 8, wherein the reverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

(228) Clause 10.

(229) The semiconductor module according to any of clauses 2 to 9, further comprising a second conductive bonding member provided between the conductive substrate and the semiconductor element to electrically bond the conductive substrate and the semiconductor element, wherein the semiconductor element has an element obverse surface facing the same side as the obverse surface, an element reverse surface facing an opposite side from the element obverse surface, and a reverse surface electrode on the element reverse surface, and wherein the second conductive bonding member includes a second base layer made of metal, a third layer provided between the second base layer and the semiconductor element and bonded in direct contact with the reverse-surface

electrode at a bonding interface with the reverse-surface electrode, and a fourth layer provided between the second base layer and the conductive substrate and bonded in direct contact with the conductive substrate at a bonding interface with the conductive substrate.

Clause 11.

(230) The semiconductor module according to clause 10, wherein a Young's modulus of the second base layer is smaller than a Young's modulus of each of the third layer and the fourth layer.

(231) Clause 12.

(232) The semiconductor module according to clause 11, wherein the second base layer is thicker than each of the third layer and the fourth layer.

(233) Clause 13.

(234) The semiconductor module according to clause 12, wherein the second base layer is made of a sheet material containing aluminum, and each of the third layer and the fourth layer is a silver plating layer stacked on a surface of the second base layer.

Clause 14.

(235) The semiconductor module according to any of clauses 10 to 13, wherein the conductive substrate has an obverse-surface bonding layer made of metal that includes the obverse surface and is stacked on a surface of the base member in the one sense of the thickness direction, and the obverse-surface bonding layer and the fourth layer are bonded in direct contact with each other at a bonding interface.

Clause 15.

(236) The semiconductor module according to clause 14, wherein the obverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

(237) Clause 16.

(238) The semiconductor module according to any of clauses 2 to 15, wherein the conductive substrate includes a first conductive portion and a second conductive portion that are spaced apart from each other in a first direction perpendicular to the thickness direction, such that the first conductive portion is offset in one sense of the first direction and the second conductive portion is offset in another sense of the first direction, the semiconductor element includes a plurality of first semiconductor elements and a plurality of second semiconductor elements, the first semiconductor elements being electrically bonded to the first conductive portion and arranged at intervals along a second direction perpendicular to the thickness direction and the first direction, the plurality of semiconductor elements being electrically bonded to the second conductive portion and arranged at intervals along the second direction, the input terminal includes a first input terminal and a second input terminal, the first input terminal being offset in the one sense of the first direction relative to the plurality of first semiconductor elements and electrically connected to the first conductive portion, the second input terminal being offset in the one sense of the first direction relative to the plurality of first semiconductor elements and electrically connected to the plurality of second semiconductor elements, and the output terminal is offset in the other sense of the first direction relative to the plurality of second semiconductor elements and electrically connected to the second conductive portion.

Clause 17.

(239) The semiconductor module according to clause 16, wherein the plurality of first semiconductor elements and the plurality of second semiconductor elements overlap with each other as viewed in the first direction.

(240) Clause 18.

(241) A semiconductor module comprising: a supporting substrate having a supporting surface that faces in one sense of a thickness direction; a conductive substrate having an obverse surface and a reverse surface that face away from each other in the thickness direction, the conductive substrate being supported by the supporting substrate such that the reverse surface faces the supporting surface; a first conductive bonding member provided between the supporting substrate and the

conductive substrate and electrically bonding the supporting substrate and the conductive substrate; and a semiconductor element electrically bonded to the obverse surface and having a switching function, wherein the first conductive bonding member includes a first base layer made of metal, a first layer, and a second layer, the first layer being provided between the first base layer and the conductive substrate and bonded to the conductive substrate by solid-phase diffusion of metal, the second layer being provided between the first base layer and the supporting substrate and bonded to the supporting substrate by solid-phase diffusion of metal.

Clause 19.

(242) The semiconductor module according to clause 18, wherein a Young's modulus of the first base layer is smaller than a Young's modulus of each of the first layer and the second layer.

(243) Clause 20.

(244) The semiconductor module according to clause 19, wherein the first base layer is thicker than each of the first layer and the second layer.

(245) Clause 21.

(246) The semiconductor module according to clause 20, wherein each of the first layer and the second layer contains silver.

(247) Clause 22.

(248) The semiconductor module according to clause 21, wherein the first base layer is made of a sheet material containing aluminum, and each of the first layer and the second layer is a silver plating layer stacked on a surface of the first base layer.

Clause 23.

(249) The semiconductor module according to any of clauses 18 to 22, wherein the supporting substrate has an insulating layer, a first metal layer stacked on a surface of the insulating layer in the one sense of the thickness direction, and a first bonding layer made of metal that includes the supporting surface and is stacked on the first metal layer, and the first bonding layer and the second layer are bonded by solid-phase diffusion.

Clause 24.

(250) The semiconductor module according to any of clauses 18 to 23, the conductive substrate has a base member made of a metal plate, and a reverse-surface bonding layer made of metal that includes the reverse surface and is stacked on a surface of the base member in another sense of the thickness direction, and the reverse-surface bonding layer and the first layer are bonded by solid-phase diffusion.

Clause 25.

(251) The semiconductor module according to clause 24, wherein the reverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

(252) Clause 26.

(253) The semiconductor module according to any of clauses 18 to 25, further comprising a second conductive bonding member provided between the conductive substrate and the semiconductor element to electrically bond the conductive substrate and the semiconductor element, wherein the semiconductor element has an element obverse surface facing the same side as the obverse surface, an element reverse surface facing an opposite side from the element obverse surface, and a reverse surface electrode on the element reverse surface, and the second conductive bonding member includes a second base layer made of metal, a third layer provided between the second base layer and the semiconductor element and bonded to the reverse-surface electrode by solid-phase diffusion of metal, and a fourth layer provided between the second base layer and the conductive substrate and bonded to the conductive substrate by solid-phase diffusion of metal.

Clause 27.

(254) The semiconductor module according to clause 26, wherein a Young's modulus of the second base layer is smaller than a Young's modulus of each of the third layer and the fourth layer.

(255) Clause 28.

(256) The semiconductor module according to clause 27, wherein the second base layer is thicker than each of the third layer and the fourth layer.

(257) Clause 29.

(258) The semiconductor module according to clause 28, wherein the second base layer is made of a sheet material containing aluminum, and each of the third layer and the fourth layer is a silver plating layer stacked on a surface of the second base layer.

Clause 30.

(259) The semiconductor module according to any of clauses 26 to 29, wherein the conductive substrate has an obverse-surface bonding layer made of metal that includes the obverse surface and is stacked on a surface of the base member in the one sense of the thickness direction, and the obverse-surface bonding layer and the fourth layer are bonded by solid-phase diffusion.

Clause 31.

(260) The semiconductor module according to clause 30, wherein the obverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

(261) Clause 32.

(262) The semiconductor module according to any of clauses 18 to 31, wherein the conductive substrate includes a first conductive portion and a second conductive portion that are spaced apart from each other as viewed in the thickness direction, the first conductive portion being offset in one sense of a first direction that is perpendicular to the thickness direction, the second conductive portion being offset in another sense of the first direction, and the semiconductor element includes a plurality of first semiconductor elements and a plurality of second semiconductor elements, the first semiconductor elements being electrically bonded to the first conductive portion and arranged at intervals along a second direction perpendicular to the thickness direction and the first direction, the plurality of semiconductor elements being electrically bonded to the second conductive portion and arranged at intervals along the second direction, the semiconductor module further comprising: a first input terminal that is offset in the one sense of the first direction relative to the plurality of first semiconductor elements and connected to the first conductive portion; a second input terminal that is offset in the one sense of the first direction relative to the plurality of first semiconductor elements and connected to the plurality of second semiconductor elements; and an output terminal that is offset in the other sense of the first direction relative to the plurality of second semiconductor elements and connected to the second conductive portion.

Clause 33.

(263) The semiconductor module according to clause 32, wherein the plurality of first semiconductor elements and the plurality of second semiconductor elements overlap with each other as viewed in the first direction.

(264) TABLE-US-00001 REFERENCE NUMERALS A1, A2, A3: Semiconductor Module 10: Semiconductor Element 10A: First Semiconductor Element 10B: Second Semiconductor Element 101: Element Obverse Surface 102: Element Reverse Surface 11: First Obverse-Surface Electrode (Gate Electrode) 12: Second Obverse-Surface Electrode (Source Electrode) 13: Third Obverse-Surface Electrode 14: Fourth Obverse-Surface Electrode 15: Reverse-Surface Electrode (Drain Electrode) 16: Fifth Obverse-Surface Electrode 171, 172, 173, 174, 181, 182, 183, 184: Corner 191: First Side 192: Second Side 193: Third Side 194: Fourth Side 2: Conductive Substrate 2A: First Conductive Portion 2B: Second Conductive Portion 201: Obverse Surface 201a: Recessed Portion 201b: Recess Edge 202: Reverse Surface 21: Base Member 22: Obverse-Surface Bonding Layer 23: Reverse-Surface Bonding Layer 3: Supporting Substrate 301: Supporting Surface 302: Bottom Surface 31: Insulating Layer 32: First Metal Layer 32A: First Portion 32B: Second Portion 321: First Bonding Layer 33: Second Metal Layer 41: Input Terminal (First Input Terminal) 411: Input-Side Bonding Surface 412: Input-Side Side Surface 413: Tip Surface 414: Lateral Surface 42: Second Input Terminal 421: Input-Side Bonding Surface 422: Input-Side Side Surface 423: Tip Surface 424: Lateral Surface 43: Third Input Terminal 431: Input-Side Bonding Surface 432: Input-

Side Surface 433: Tip Surface 434: Lateral Surface 44: Output Terminal 441: Output-Side Bonding Surface 442: Output-Side Side Surface 443: Tip Surface 444: Lateral Surface 45: Control Terminal 451: Holder 452: Metal Pin 459: Conductive Bonding Member 46A, 46B, 46C, 46D, 46E: First Control Terminal 47A, 47B, 47C, 47D: Second Control Terminal 5: Control Terminal Support 51: Insulating Layer 52: First Metal Layer 521: First Portion 522: Second Portion 523: Third Portion 524: Fourth Portion 525: Fifth Portion 53: Second Metal Layer 59: Bonding Member 6: Conducting Member 601: First Portion 61: First Conducting Member 61h: Opening 62: Second Conducting Member 62A: First Portion 62B: Second Portion 621: First Wiring Portion 622: Second Wiring Portion 623: Third Wiring Portion 623a: Recessed Area 623h: Opening 624: Fourth Wiring Portion 625: First Band portion 625a: Protruding Area 625h: Opening 626: Second Band portion 627: First Edge 628: Second Edge 63: Opening 69: Conductive Bonding Member 71: First Conductive Bonding Member 711: First Base Layer 712: First Layer 713: Second Layer 72: Second Conductive Bonding Member 721: Second Base Layer 722: Third Layer 723: Fourth Layer 731: Wire 731a: First Wire 731b: Second Wire 732, 733, 734, 735: Wire 8: Sealing Resin 81: Resin Obverse Surface 82: Resin Reverse Surface 831, 832: Resin Side Surface 832a: Recessed Portion 833, 834: Resin Side Surface 851: First Protrusion 851a: First Protruding End Surface 851b: Recessed Portion 851c: Inner Wall Surface 852: Second Protrusion 86: Resin Void 861: Resin Void Edge 87: Resin Member 88: Resin-Filling Portion 91: Mold 911: Pressing Pin

Claims

1. A semiconductor module comprising: a supporting substrate having a supporting surface that faces in one sense of a thickness direction; a conductive substrate having an obverse surface and a reverse surface that face away from each other in the thickness direction, the conductive substrate being supported by the supporting substrate such that the reverse surface faces the supporting surface; a first conductive bonding member provided between the supporting substrate and the conductive substrate and electrically bonding the supporting substrate and the conductive substrate; and a semiconductor element electrically bonded to the obverse surface and having a switching function, wherein the first conductive bonding member includes a first base layer made of metal, a first layer, and a second layer, the first layer being provided between the first base layer and the conductive substrate and bonded in direct contact with the conductive substrate at a bonding interface with the conductive substrate, the second layer being provided between the first base layer and the supporting substrate and bonded in direct contact with the supporting substrate at a bonding interface with the supporting substrate.
2. The semiconductor module according to claim 1, further comprising: a sealing resin having a resin obverse surface facing the same side as the obverse surface, a resin reverse surface facing an opposite side from the resin obverse surface, and a resin side surface connected to the resin obverse surface and the resin reverse surface and flanked by the resin obverse surface and the resin reverse surface in the thickness direction, the sealing resin covering at least a part of the supporting substrate, the conductive substrate, the first conductive bonding member, and the semiconductor element; an input terminal electrically connected to the semiconductor element and protruding from the resin side surface; an output terminal electrically connected to the semiconductor element, and protruding from the resin side surface; and a control terminal for controlling the semiconductor element, wherein the input terminal has an input-side bonding surface facing in the one sense of the thickness direction, the output terminal has an output-side bonding surface facing in the one sense of the thickness direction, and the control terminal is arranged on the obverse surface and extends along the thickness direction.
3. The semiconductor module according to claim 2, wherein a Young's modulus of the first base layer is smaller than a Young's modulus of each of the first layer and the second layer.
4. The semiconductor module according to claim 3, wherein the first base layer is thicker than each

of the first layer and the second layer.

5. The semiconductor module according to claim 4, wherein each of the first layer and the second layer contains silver.

6. The semiconductor module according to claim 5, wherein the first base layer is made of a sheet material containing aluminum, and each of the first layer and the second layer is a silver plating layer stacked on a surface of the first base layer.

7. The semiconductor module according to claim 2, wherein the supporting substrate has an insulating layer, a first metal layer stacked on a surface of the insulating layer in the one sense of the thickness direction, and a first bonding layer made of metal that includes the supporting surface and is stacked on the first metal layer, and the first bonding layer and the second layer are bonded in direct contact with each other at a bonding interface.

8. The semiconductor module according to claim 2, wherein the conductive substrate has a base member made of a metal plate, and a reverse-surface bonding layer made of metal that includes the reverse surface and is stacked on a surface of the base member in another sense of the thickness direction, and the reverse-surface bonding layer and the first layer are bonded in direct contact with each other at a bonding interface.

9. The semiconductor module according to claim 8, wherein the reverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

10. The semiconductor module according to claim 2, further comprising a second conductive bonding member provided between the conductive substrate and the semiconductor element to electrically bond the conductive substrate and the semiconductor element, wherein the semiconductor element has an element obverse surface facing the same side as the obverse surface, an element reverse surface facing an opposite side from the element obverse surface, and a reverse surface electrode on the element reverse surface, and wherein the second conductive bonding member includes a second base layer made of metal, a third layer provided between the second base layer and the semiconductor element and bonded in direct contact with the reverse-surface electrode at a bonding interface with the reverse-surface electrode, and a fourth layer provided between the second base layer and the conductive substrate and bonded in direct contact with the conductive substrate at a bonding interface with the conductive substrate.

11. The semiconductor module according to claim 10, wherein a Young's modulus of the second base layer is smaller than a Young's modulus of each of the third layer and the fourth layer.

12. The semiconductor module according to claim 11, wherein the second base layer is thicker than each of the third layer and the fourth layer.

13. The semiconductor module according to claim 12, wherein the second base layer is made of a sheet material containing aluminum, and each of the third layer and the fourth layer is a silver plating layer stacked on a surface of the second base layer.

14. The semiconductor module according to claim 10, wherein the conductive substrate has an obverse-surface bonding layer made of metal that includes the obverse surface and is stacked on a surface of the base member in the one sense of the thickness direction, and the obverse-surface bonding layer and the fourth layer are bonded in direct contact with each other at a bonding interface.

15. The semiconductor module according to claim 14, wherein the obverse-surface bonding layer is a silver plating layer stacked on the surface of the base member.

16. The semiconductor module according to claim 2, wherein the conductive substrate includes a first conductive portion and a second conductive portion that are spaced apart from each other in a first direction perpendicular to the thickness direction, such that the first conductive portion is offset in one sense of the first direction and the second conductive portion is offset in another sense of the first direction, the semiconductor element includes a plurality of first semiconductor elements and a plurality of second semiconductor elements, the first semiconductor elements being electrically bonded to the first conductive portion and arranged at intervals along a second direction

perpendicular to the thickness direction and the first direction, the plurality of semiconductor elements being electrically bonded to the second conductive portion and arranged at intervals along the second direction, the input terminal includes a first input terminal and a second input terminal, the first input terminal being offset in the one sense of the first direction relative to the plurality of first semiconductor elements and electrically connected to the first conductive portion, the second input terminal being offset in the one sense of the first direction relative to the plurality of first semiconductor elements and electrically connected to the plurality of second semiconductor elements, and the output terminal is offset in the other sense of the first direction relative to the plurality of second semiconductor elements and electrically connected to the second conductive portion.

17. The semiconductor module according to claim 16, wherein the plurality of first semiconductor elements and the plurality of second semiconductor elements overlap with each other as viewed in the first direction.
